

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	2
1.1	DataFrame import and filters	2
1.2	Metadata	4
2	Model(s) analysis	5
2.1	Explanation legend tables	5
2.2	Repartition of train and test	6
2.2.1	External Split: X_train vs X_test	6
2.2.2	Internal CV Folds	6
2.3	Model : Linear	8
2.3.1	Gridsearch	8
2.3.2	Scatter	10
2.3.3	Scatter residuals	11
2.3.4	Learning curve	11
2.4	Model : LGBM	12
2.4.1	Gridsearch	12
2.4.2	Scatter	17
2.4.3	Scatter residuals	18
2.4.4	Learning curve	18
2.5	Model : CatBoost	19
2.5.1	Gridsearch	19
2.5.2	Scatter	25
2.5.3	Scatter residuals	26
2.5.4	Learning curve	26
2.6	Sumup-table	27
3	Plot	27
3.1	Standalone plots	27
3.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	27
3.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	30

3.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	33
3.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	36
3.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	39
3.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	42
3.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	45
3.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	48
3.2	Compare plots	51
3.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	51
3.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	53
3.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	54
3.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	55
3.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	57
3.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	58
3.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	59
3.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	61

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
```

```
#df['campaign'] = "ALL"
```

```
return df
```

```
def import_data_IR():
```

```
    if target_col[0] == "i_NH3":
```

```
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
```

```
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
```

```
    elif target_col[0] == "i_CO2":
```

```
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
```

```
    else:
```

```
        raise ValueError("Target is not supported")
```

```
df = pd.read_csv(file_path, sep=";")
```

```
df["datetime"] = pd.to_datetime(df["datetime"])
```

```
df.sort_values("datetime").reset_index(drop=True)
```

```
if target_col[0] == "i_NH3":
```

```
    df["campaign"] = df["datetime"].apply(assign_campaign)
```

```
    df = df[df["campaign"] != "other"].copy()
```

```
    df["innova_point"] = df.apply(assign_innova_point, axis=1)
```

```
    df = df[df["innova_point"] != "no"].copy()
```

```
    with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
```

```
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
```

```
            dict_features = json.load(f)
```

```
elif target_col[0] == "i_CO2":
```

```
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
```

```
    df = df[df["campaign"] != "other"].copy()
```

```
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
```

```
    df = df[df["innova_point"] != "no"].copy()
```

```
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
```

```
        dict_features = json.load(f)
```

```
else:
```

```
    raise ValueError("The target col do not match any existing function")
```

```
df["hot"] = df.apply(assign_hot_season,axis=1)
```

```
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
```

```
#Removing of rabbit 4
```

```
df["column"] = df.apply(assign_colonna, axis = 1)
```

```
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450

Métrique utilisée	r2
-------------------	----

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

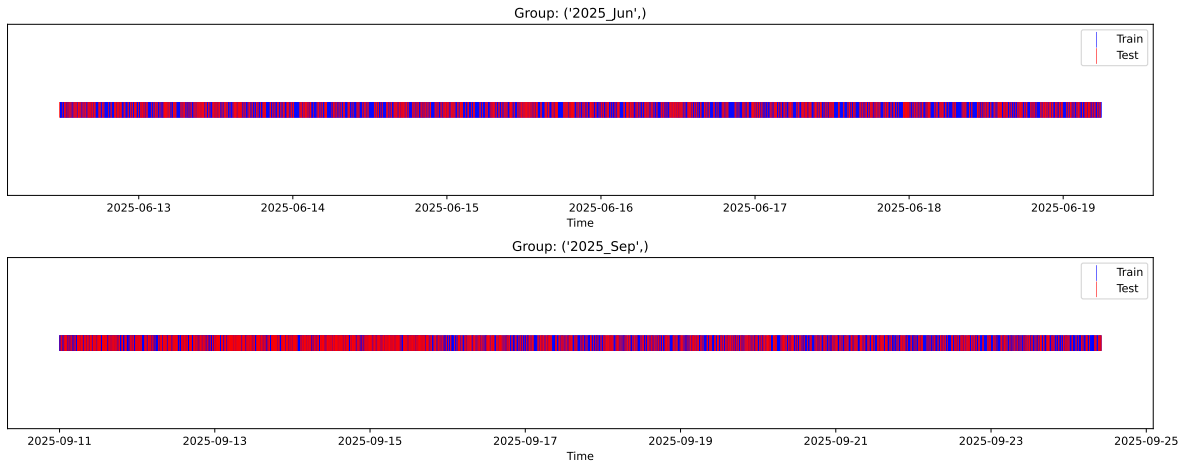
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

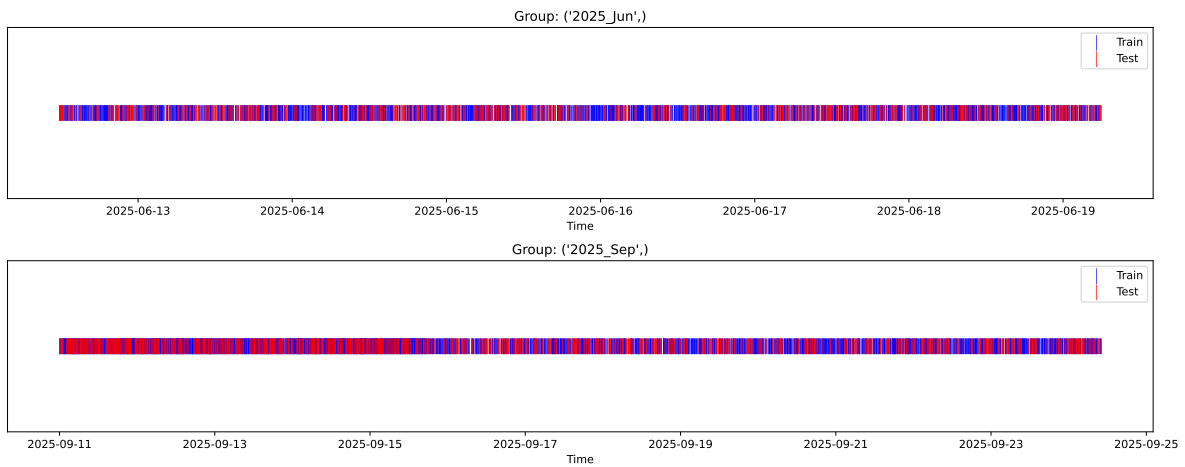
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

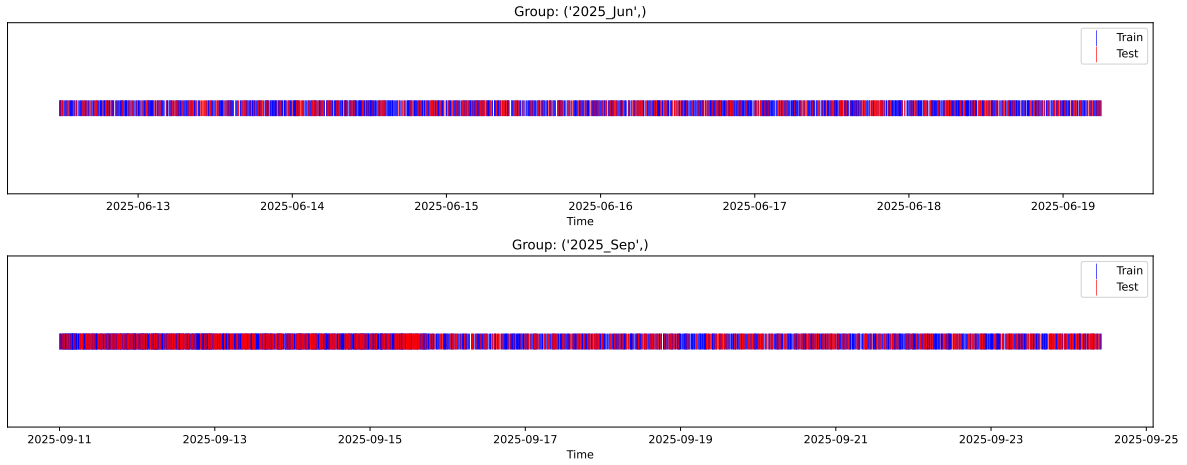
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



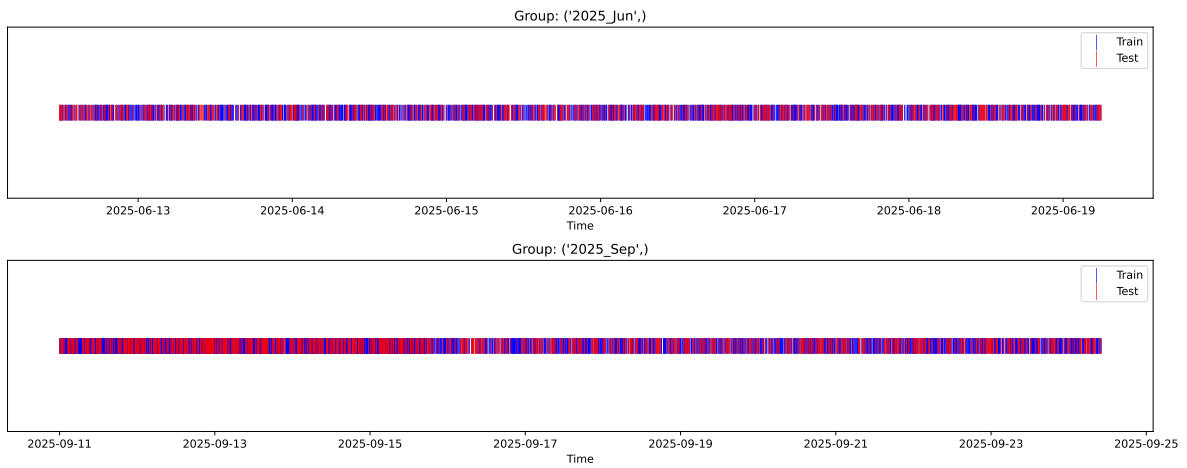
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



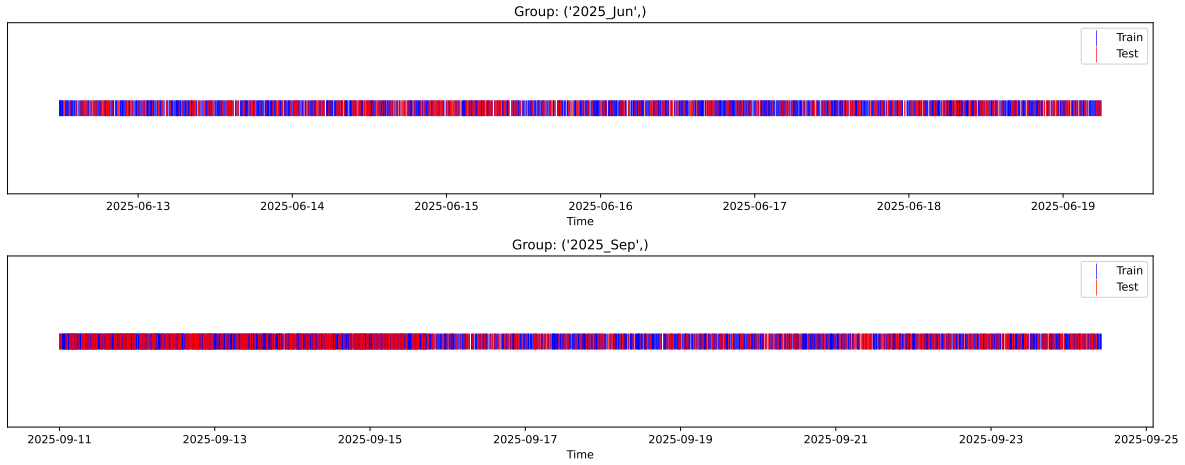
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



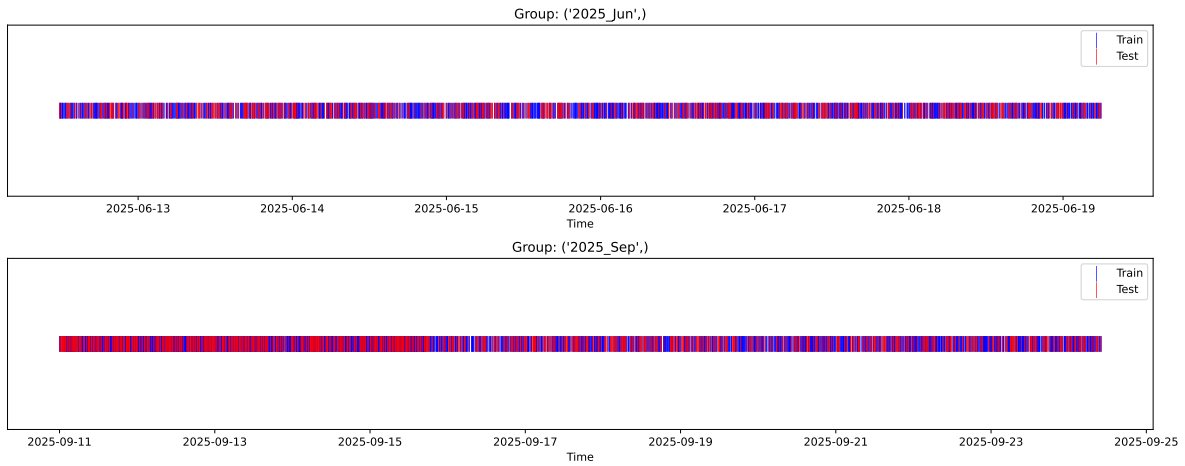
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.339, std=0.798, min=0.959, max=13.845
 y_{test} : mean=2.374, std=0.786, min=1.071, max=7.277

Fold 0 — y_{test} : mean=2.34, std=0.82, y_{train} : mean=2.34, std=0.79

Fold 1 — y_{test} : mean=2.34, std=0.77, y_{train} : mean=2.34, std=0.81

Fold 2 — y_{test} : mean=2.36, std=0.85, y_{train} : mean=2.33, std=0.78

Fold 3 — y_test: mean=2.36, std=0.84, y_train: mean=2.33, std=0.78

Fold 4 — y_test: mean=2.35, std=0.76, y_train: mean=2.34, std=0.81

train scores CV: [0.35126041 0.34129782 0.38941971 0.36572649 0.35147845]

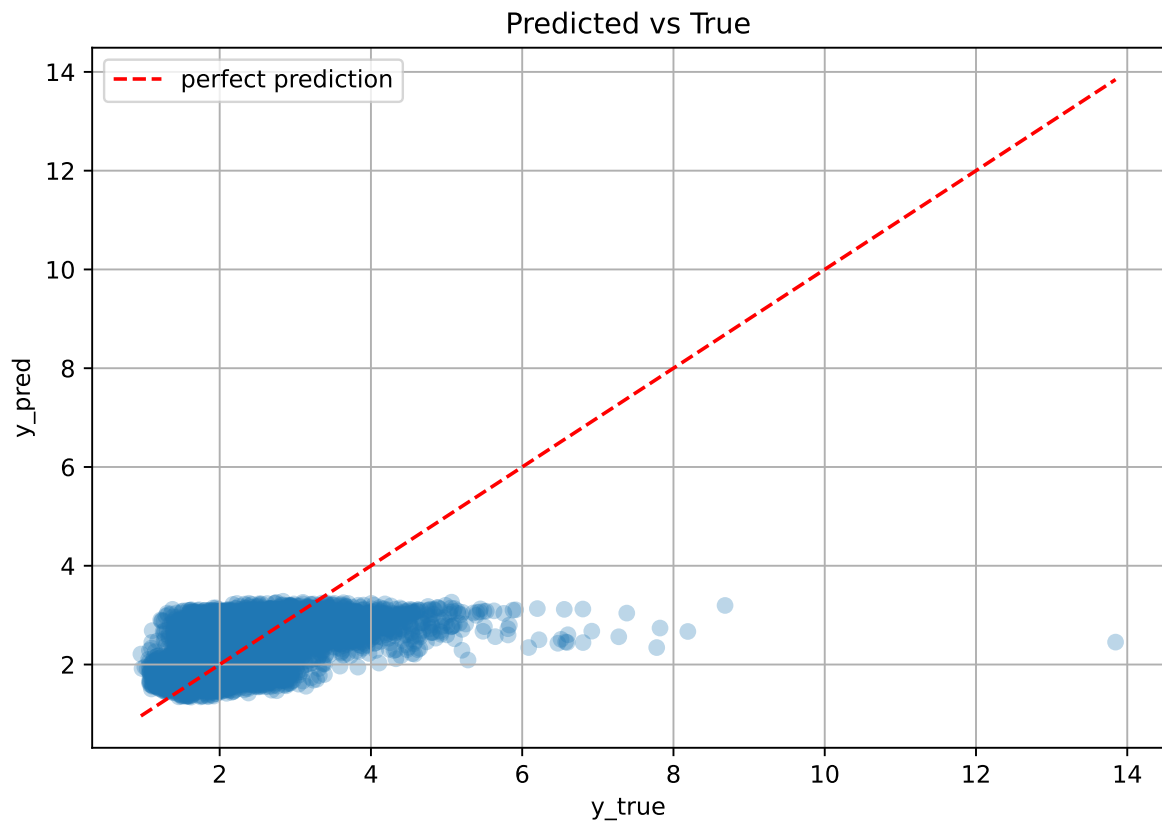
test scores CV: [0.37881413 0.40998904 0.30281638 0.34887505 0.38406497]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3598	0.3649	0.0366	0.3451	0.3453	-0.0197	0.9461	0.9861

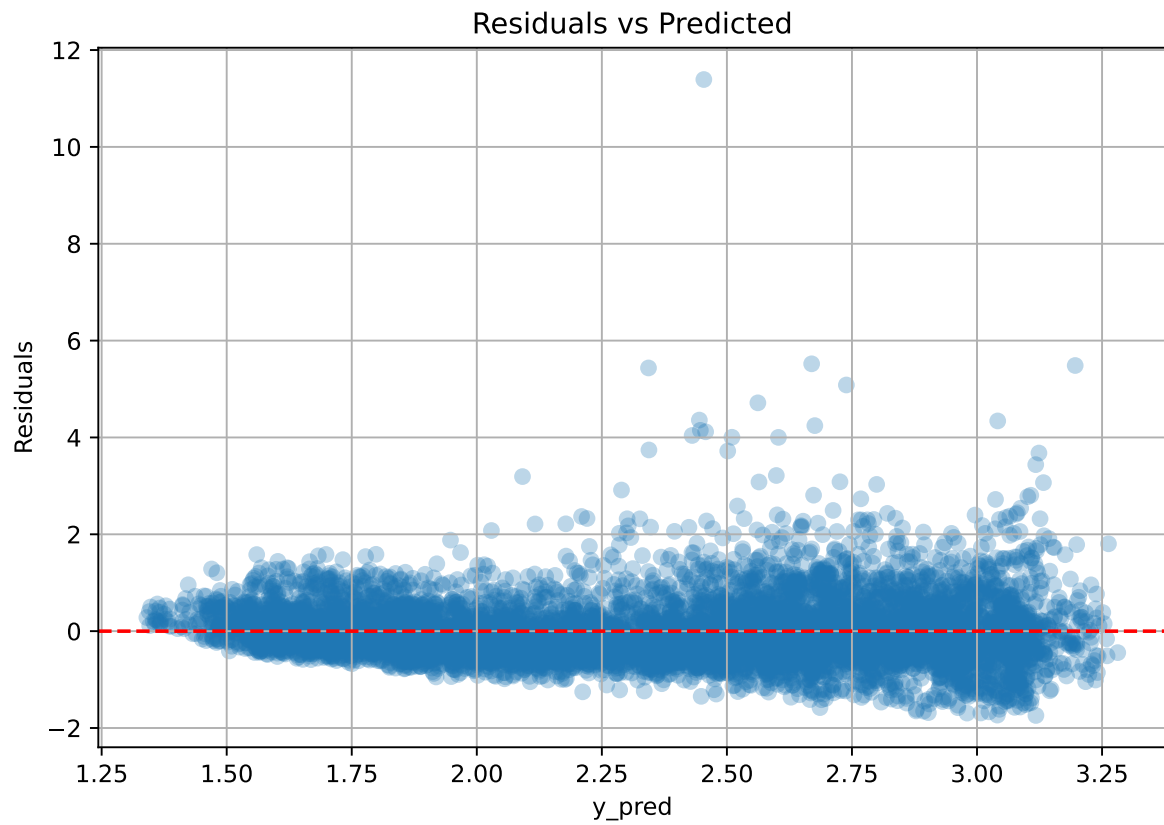
Coefficients:

coef_r_CO2 = -0.0491, coef_r_NH3 = -0.0447, coef_r_THI = 0.5970, coef_r_temperature = -0.7372, coef_r_umidity = -0.6671, intercept = 2.3392,

2.3.2 Scatter



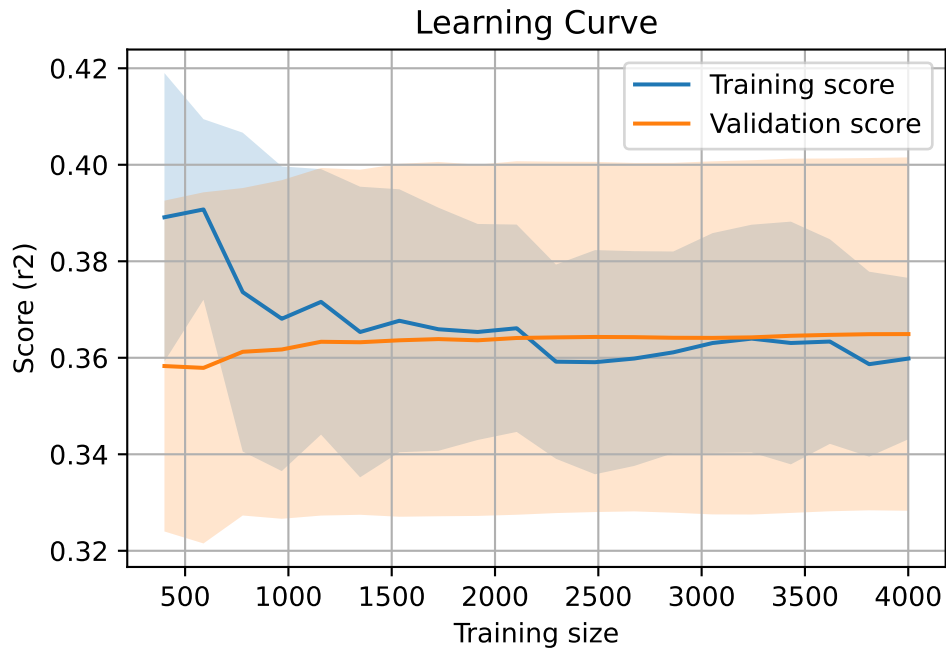
2.3.3 Scatter residuals



2.3.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

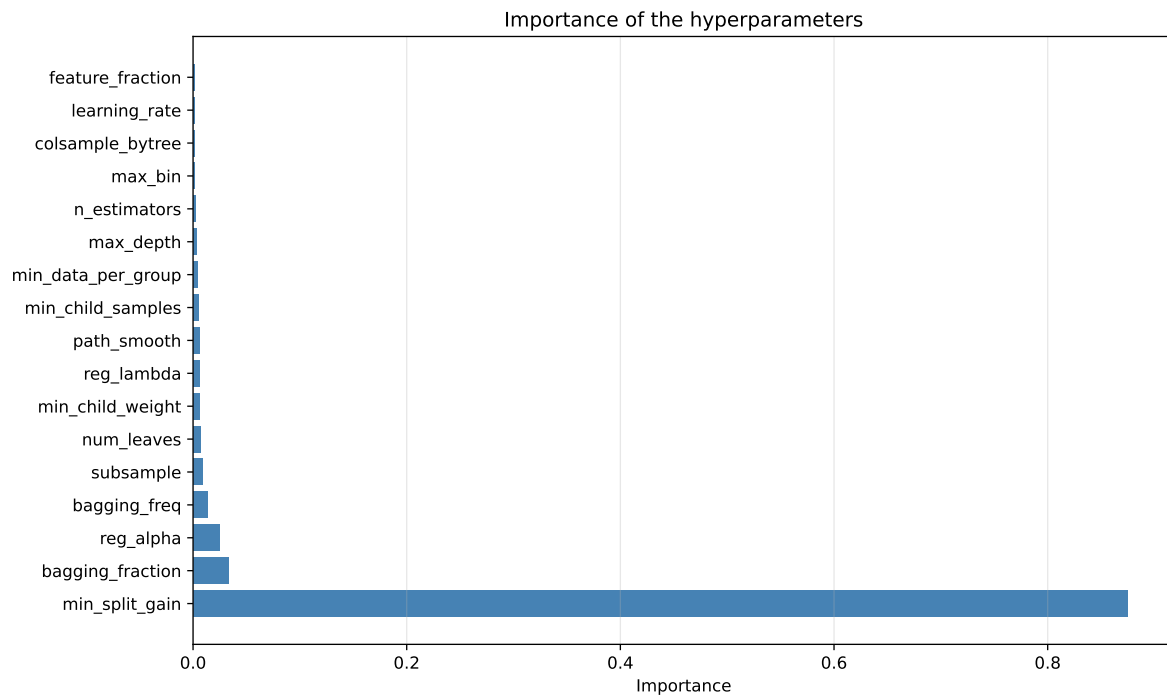
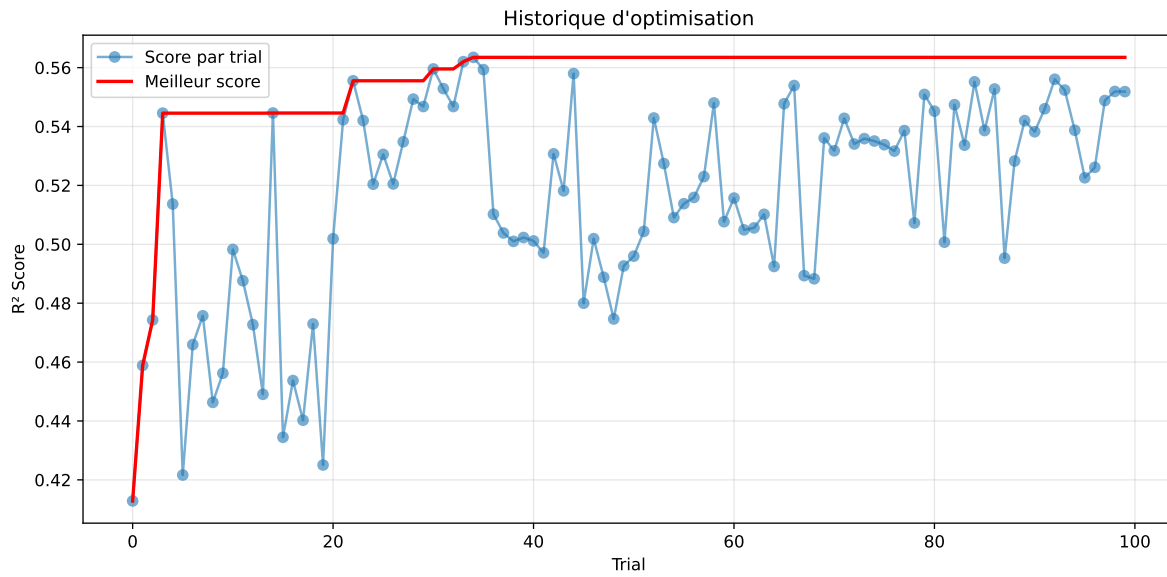


2.4 Model : LGBM

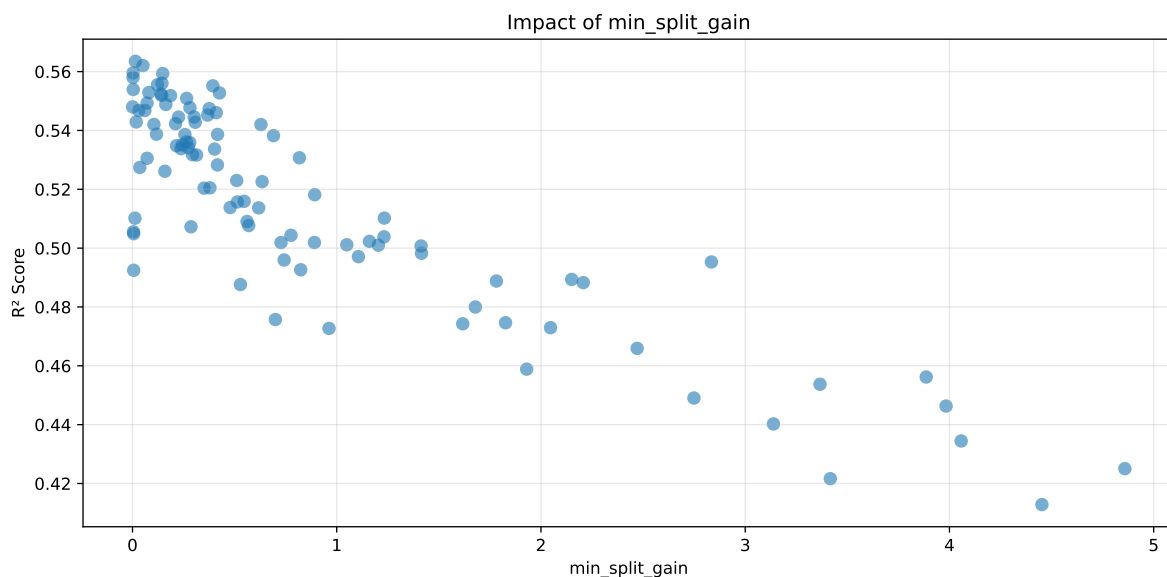
2.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845
y_test: mean=2.374, std=0.786, min=1.071, max=7.277

===== TOP Importance FEATURES =====
feature importance r_CO2 4215 r_NH3 4126 r_umidity 3740 r_THI 3517 r_temperature
2959



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6973	0.5635	0.5987	0.0394	0.5635	0.6124	0.0489	1.0869	1.2375
2	0.6966	0.562	0.5992	0.0406	0.562	0.6136	0.0516	1.0918	1.2394
3	0.7017	0.5596	0.6003	0.0393	0.5596	0.6164	0.0568	1.1016	1.254
4	0.6905	0.5593	0.5972	0.0417	0.5593	0.6105	0.0512	1.0915	1.2346
5	0.6742	0.558	0.5914	0.0418	0.558	0.6051	0.0471	1.0844	1.2082
6	0.7111	0.556	0.5914	0.0401	0.556	0.6051	0.049	1.0881	1.2788
7	0.669	0.5555	0.5897	0.0364	0.5555	0.6009	0.0454	1.0817	1.2043
8	0.6836	0.5552	0.5888	0.0402	0.5552	0.6029	0.0478	1.0861	1.2313
9	0.6546	0.5539	0.5859	0.0413	0.5539	0.5962	0.0423	1.0764	1.1817
10	0.6617	0.5529	0.5865	0.0402	0.5529	0.6006	0.0477	1.0863	1.1969
---	---	---	---	---	---	---	---	---	---
100	0.4206	0.4128	0.4126	0.0355	0.4128	0.4452	0.0324	1.0785	1.0188

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.01877354413741985, 'reg__num_leaves': 35, 'reg__subsample': 0.8824519540732422, 'reg__colsample_bytree': 0.5453734574701726, 'reg__min_child_samples': 10, 'reg__reg_alpha': 1.5727828177865288, 'reg__reg_lambda': 33.90285472801301, 'reg__min_split_gain': 0.014504902769949951, 'reg__path_smooth': 2.479518048142483, 'reg__min_child_weight': 0.31520976091868363, 'reg__feature_fraction': 0.8982957364187277, 'reg__bagging_fraction': 0.8885906731111225, 'reg__bagging_freq': 2, 'reg__max_bin': 185, 'reg__min_data_per_group': 200}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.019501279653815325, 'reg__num_leaves': 35, 'reg__subsample': 0.8950677953153287, 'reg__colsample_bytree': 0.5041423119582233, 'reg__min_child_samples': 11, 'reg__reg_alpha': 1.5307614393689857, 'reg__reg_lambda': 34.37929425356761, 'reg__min_split_gain': 0.051260219672657356, 'reg__path_smooth': 3.069406645112665, 'reg__min_child_weight': 0.006328725761601226, 'reg__feature_fraction': 0.8857892996486324, 'reg__bagging_fraction': 0.8601482833477455, 'reg__bagging_freq': 1, 'reg__max_bin': 182, 'reg__min_data_per_group': 199}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.020209115104784046, 'reg__num_leaves': 63, 'reg__subsample': 0.6637298758727264, 'reg__colsample_bytree': 0.5517172640446842, 'reg__min_child_samples': 37, 'reg__reg_alpha': 1.7806525909648399, 'reg__reg_lambda': 37.28707830706662, 'reg__min_split_gain': 0.0029408655899241243, 'reg__path_smooth': 3.297182444989105, 'reg__min_child_weight': 0.3178829949844219, 'reg__feature_fraction': 0.7067230717467505, 'reg__bagging_fraction': 0.892252469396438, 'reg__bagging_freq': 7, 'reg__max_bin': 126, 'reg__min_data_per_group': 197}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.018910616549673187, 'reg__num_leaves': 35, 'reg__subsample': 0.8920583854362558, 'reg__colsample_bytree': 0.5507248434211297, 'reg__min_child_samples': 11, 'reg__reg_alpha': 0.1152239103539447, 'reg__reg_lambda': 35.514952876809836, 'reg__min_split_gain': 0.14783314437785883, 'reg__path_smooth': 2.203814142318947, 'reg__min_child_weight': 0.3297506943454289, 'reg__feature_fraction': 0.8278571541436492, 'reg__bagging_fraction': 0.879874718603282, 'reg__bagging_freq': 2, 'reg__max_bin': 181, 'reg__min_data_per_group': 93}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.013269226374111017, 'reg__num_leaves': 42, 'reg__subsample': 0.8564423741700671, 'reg__colsample_bytree': 0.5571396513535873, 'reg__min_child_samples': 15, 'reg__reg_alpha': 0.9080893927558825, 'reg__reg_lambda': 35.062463705113885, 'reg__min_split_gain': 0.002674461270005715, 'reg__path_smooth': 3.5418802019472846, 'reg__min_child_weight': 0.013623985920107887, 'reg__feature_fraction': 0.7279996996539678, 'reg__bagging_fraction': 0.8125006102223116, 'reg__bagging_freq': 3, 'reg__max_bin': 137, 'reg__min_data_per_group': 108}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.023379662150320345, 'reg__num_leaves': 36, 'reg__subsample': 0.6530066805437313, 'reg__colsample_bytree': 0.5222850354013258, 'reg__min_child_samples': 10, 'reg__reg_alpha': 1.3757041025869161, 'reg__reg_lambda': 6.172523825384711, 'reg__min_split_gain': 0.14460711962515127, 'reg__path_smooth': 2.279174541815983, 'reg__min_child_weight': 0.3927466829175851, 'reg__feature_fraction': 0.6886032055163219, 'reg__bagging_fraction': 0.8896995491103072, 'reg__bagging_freq': 2, 'reg__max_bin': 209, 'reg__min_data_per_group': 126}

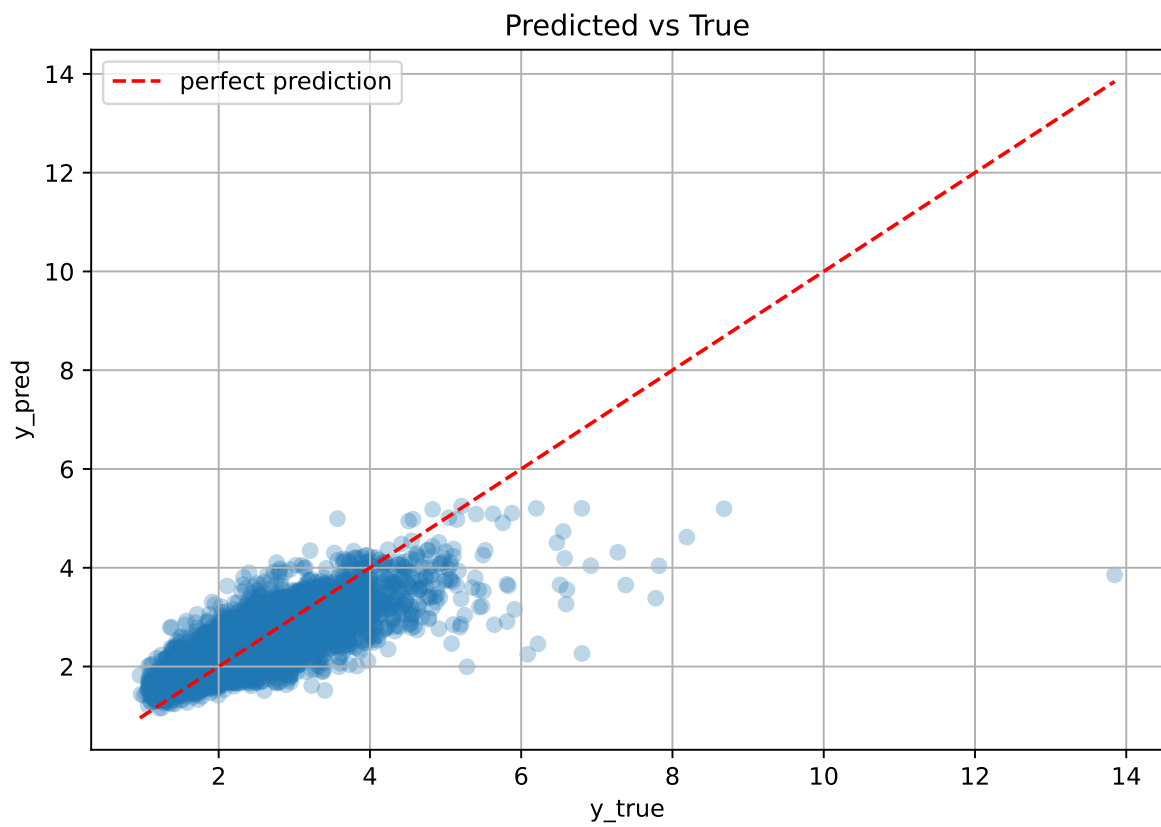
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.03481127877124342, 'reg__num_leaves': 32, 'reg__subsample': 0.8792225500981891, 'reg__colsample_bytree': 0.5326789416527989, 'reg__min_child_samples': 10, 'reg__reg_alpha': 0.22675623365964093, 'reg__reg_lambda': 0.5643931543801015, 'reg__min_split_gain': 0.12246065017494856, 'reg__path_smooth': 2.798875977176115, 'reg__min_child_weight': 0.06781963436699631, 'reg__feature_fraction': 0.8970784306272268, 'reg__bagging_fraction': 0.8113818961996091, 'reg__bagging_freq': 1, 'reg__max_bin': 64, 'reg__min_data_per_group': 198}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.011446590735549524, 'reg__num_leaves': 37, 'reg__subsample': 0.8549745562289676, 'reg__colsample_bytree': 0.7423978015240553, 'reg__min_child_samples': 10, 'reg__reg_alpha': 0.259188203149312, 'reg__reg_lambda': 6.284192452451524, 'reg__min_split_gain': 0.39340647292525555, 'reg__path_smooth': 1.0258841037690611, 'reg__min_child_weight': 0.02284331463687809, 'reg__feature_fraction': 0.7453856999006305, 'reg__bagging_fraction': 0.8544516263687949, 'reg__bagging_freq': 4, 'reg__max_bin': 219, 'reg__min_data_per_group': 82}

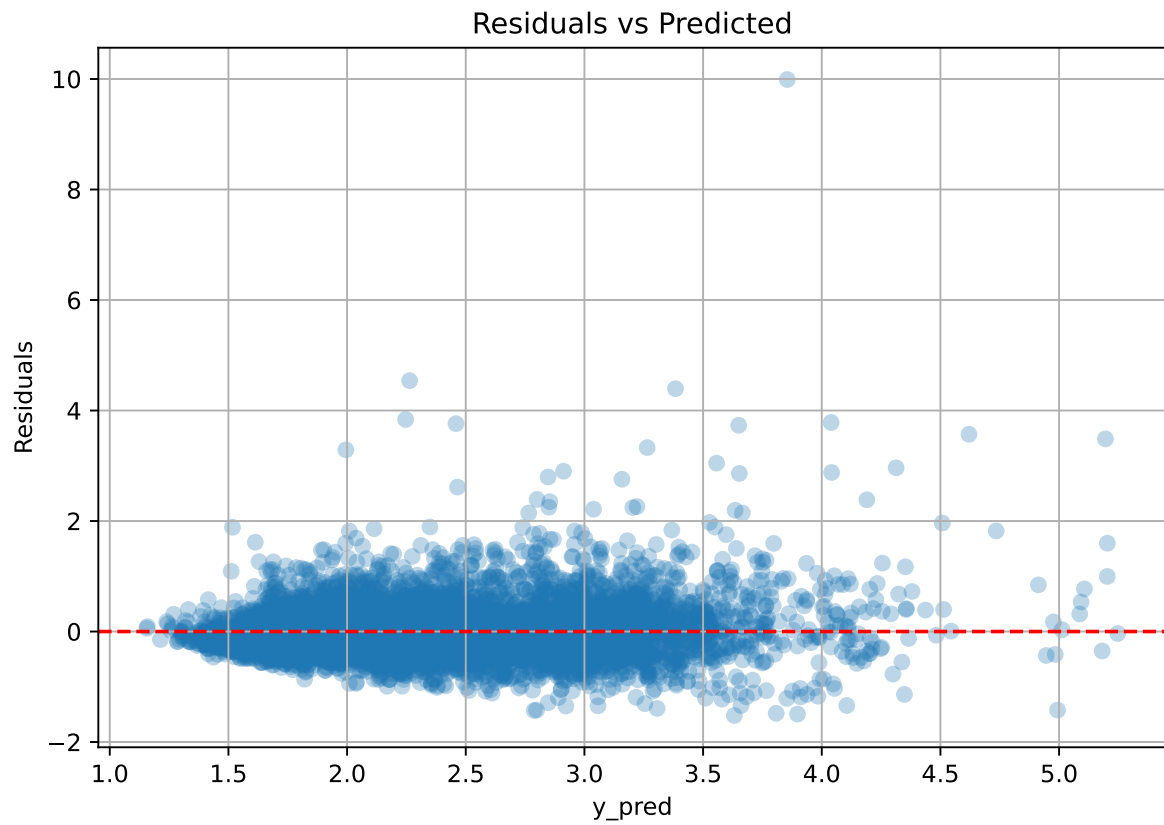
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.016487383584574324, 'reg__num_leaves': 32, 'reg__subsample': 0.879002807193783, 'reg__colsample_bytree': 0.6501762771349126, 'reg__min_child_samples': 19, 'reg__reg_alpha': 0.606788457033218, 'reg__reg_lambda': 25.81877066061692, 'reg__min_split_gain': 0.0036074797787762347, 'reg__path_smooth': 1.6208074457171462, 'reg__min_child_weight': 0.0036905742427814745, 'reg__feature_fraction': 0.8667238531043621, 'reg__bagging_fraction': 0.8998109507541605, 'reg__bagging_freq': 2, 'reg__max_bin': 196, 'reg__min_data_per_group': 141}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.02038863705155775, 'reg__num_leaves': 35, 'reg__subsample': 0.6663326260996643, 'reg__colsample_bytree': 0.5493012128014735, 'reg__min_child_samples': 11, 'reg__reg_alpha': 1.719653933261229, 'reg__reg_lambda': 32.992757744698466, 'reg__min_split_gain': 0.08015698300205432, 'reg__path_smooth': 3.0595229794252936, 'reg__min_child_weight': 0.6097009870911929, 'reg__feature_fraction': 0.7199723398259904, 'reg__bagging_fraction': 0.8285619296085102, 'reg__bagging_freq': 7, 'reg__max_bin': 188, 'reg__min_data_per_group': 199}

2.4.2 Scatter



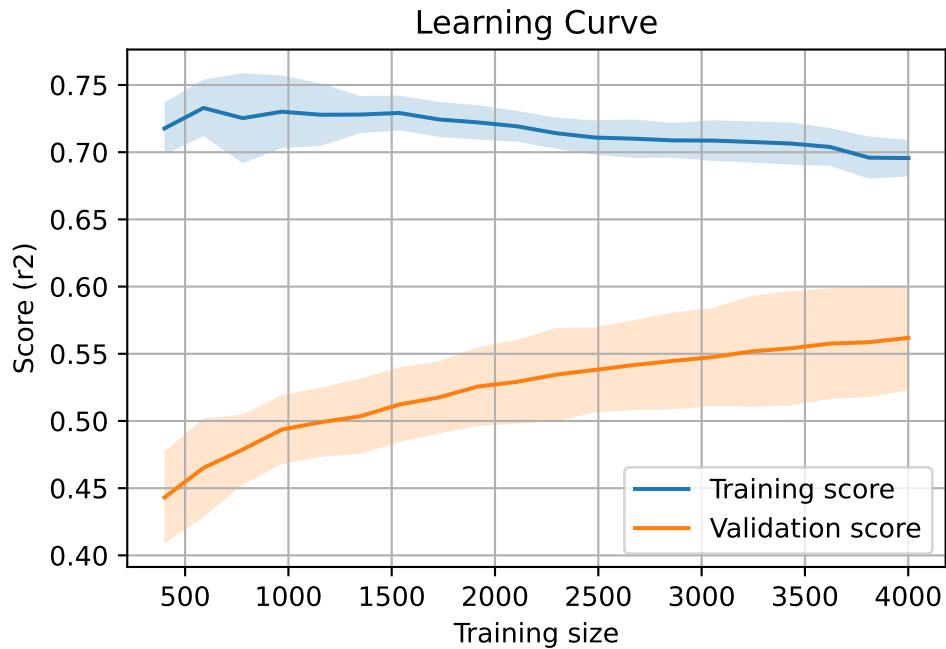
2.4.3 Scatter residuals



2.4.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

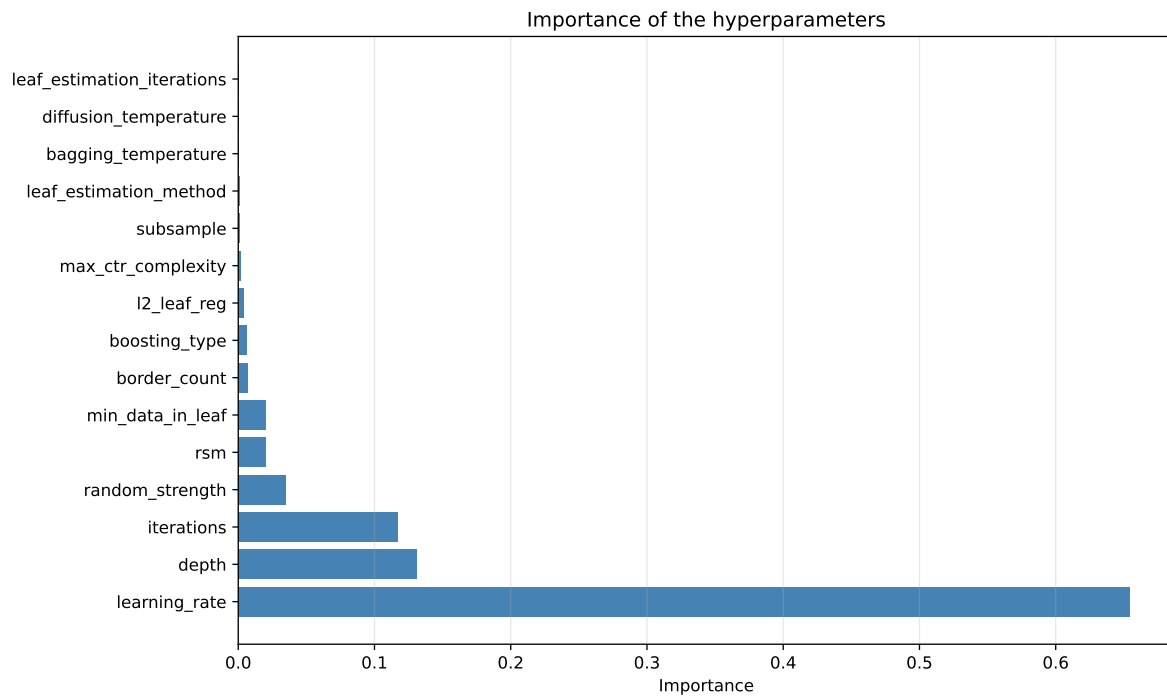
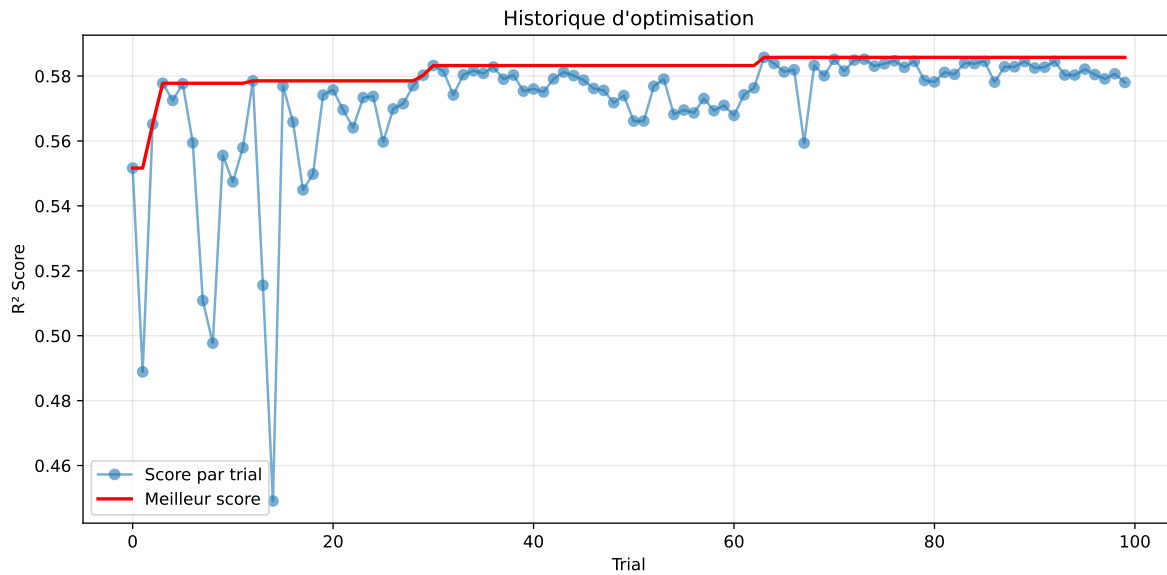


2.5 Model : CatBoost

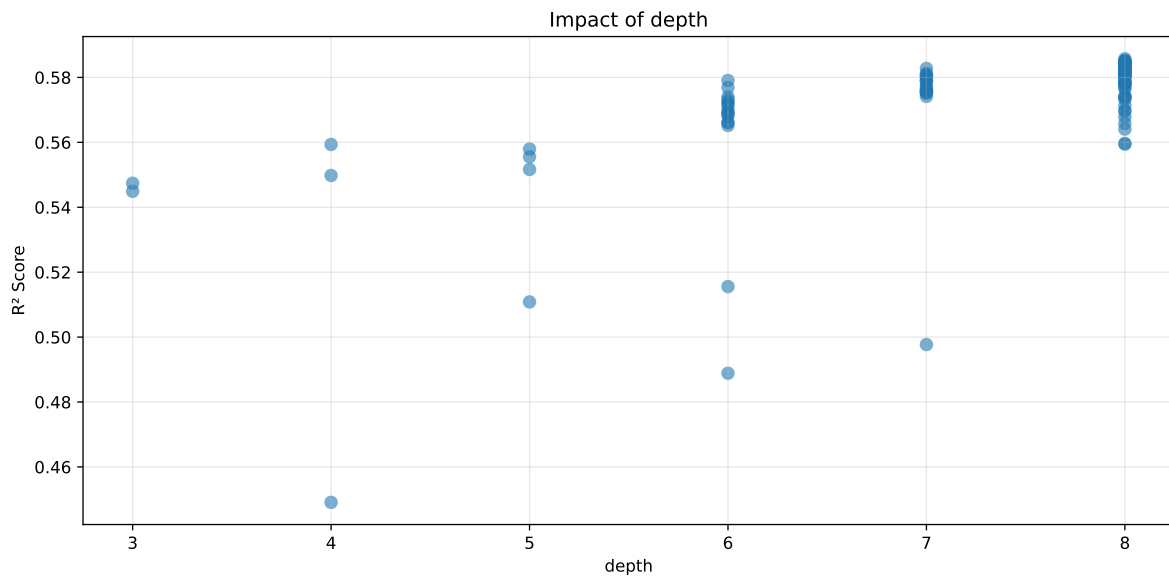
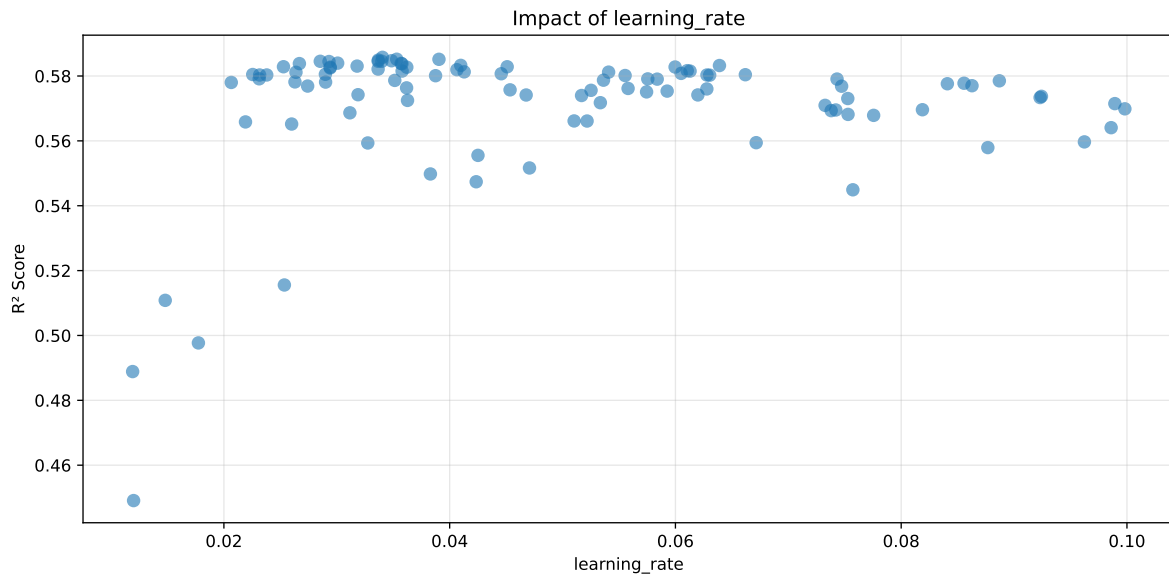
2.5.1 Gridsearch

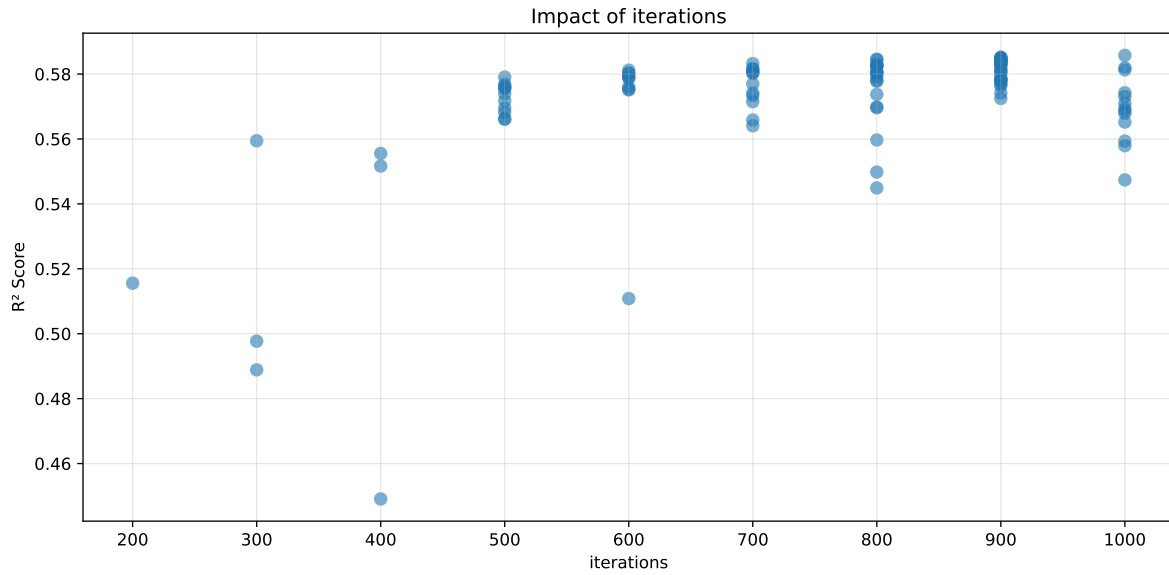
Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845
y_test: mean=2.374, std=0.786, min=1.071, max=7.277

===== TOP Importance FEATURES =====
feature importance r_NH3 26.830345 r_umidity 26.217432 r_CO2 22.071906 r_THI 14.316489
r_temperature 10.563827



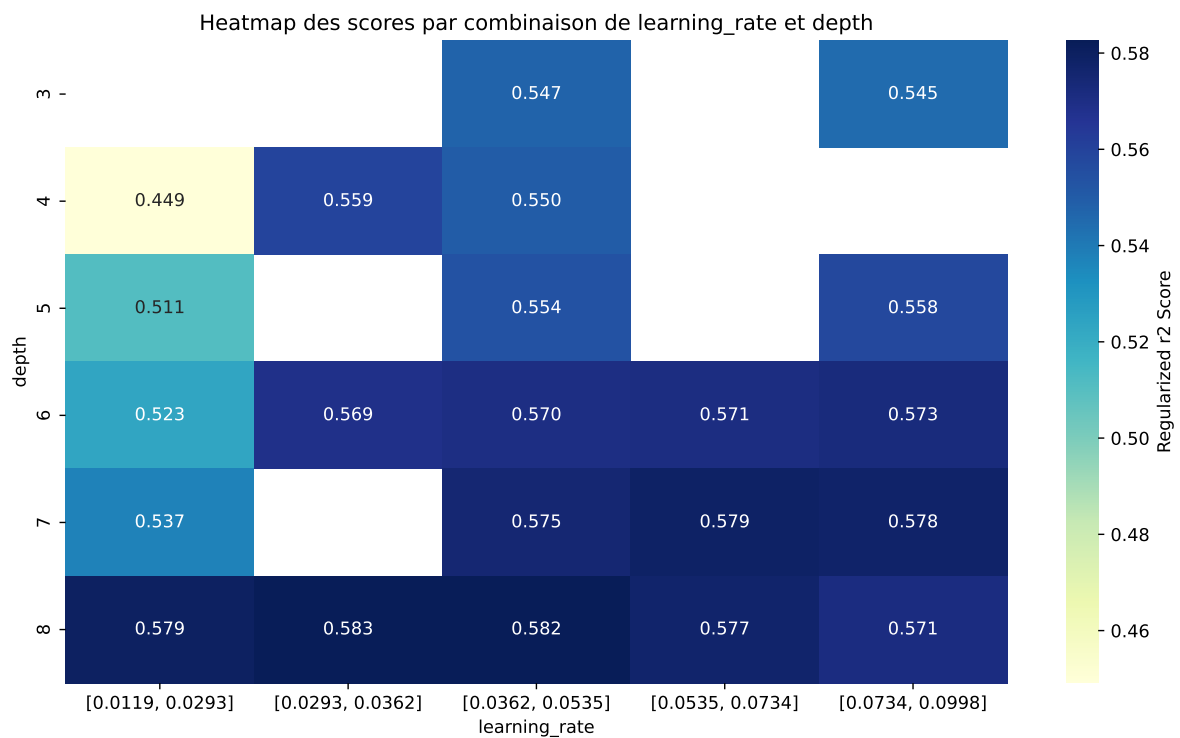
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.8363	0.5858	0.6237	0.0395	0.5858	0.6382	0.0524	1.0895	1.4278
2	0.8472	0.5852	0.6233	0.0371	0.5852	0.636	0.0508	1.0868	1.4479
3	0.8575	0.5851	0.6244	0.0382	0.5851	0.6363	0.0511	1.0874	1.4655
4	0.8356	0.5849	0.622	0.0359	0.5849	0.6375	0.0526	1.0899	1.4286
5	0.8292	0.5847	0.6232	0.038	0.5847	0.6351	0.0504	1.0861	1.4181
6	0.8103	0.5845	0.6211	0.0357	0.5845	0.6379	0.0534	1.0913	1.3862
7	0.8161	0.5845	0.6214	0.0353	0.5845	0.6369	0.0523	1.0895	1.3962
8	0.8081	0.5845	0.6221	0.0361	0.5845	0.6346	0.0501	1.0857	1.3824
9	0.7848	0.5845	0.6183	0.0372	0.5845	0.6282	0.0438	1.0749	1.3429
10	0.8207	0.584	0.6219	0.0377	0.584	0.637	0.053	1.0908	1.4054
---	---	---	---	---	---	---	---	---	---
100	0.4585	0.4491	0.4588	0.0369	0.4491	0.4641	0.015	1.0335	1.021

Hyperparameters:

Rank 1: {'reg__iterations': 1000, 'reg__depth': 8, 'reg__learning_rate': 0.03404600355065968, 'reg__l2_leaf_reg': 28.096501195995316, 'reg__bagging_temperature': 2.3898930213586165, 'reg__random_strength': 3.6512227494988827, 'reg__subsample': 0.5253738582547043, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8281765825822307, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9230.124246103598}

Rank 2: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.035310686389678575, 'reg__l2_leaf_reg': 12.181002497942284, 'reg__bagging_temperature': 3.5367776009214493, 'reg__random_strength': 2.700003125882999, 'reg__subsample': 0.5010691163970088, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8252835936319267, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8463.04630811414}

Rank 3: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.039051455968782615, 'reg__l2_leaf_reg': 14.683192578821739, 'reg__bagging_temperature': 2.4529355998016116, 'reg__random_strength': 2.73601001585431, 'reg__subsample': 0.5272883240974628, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8251146805761697, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8767.00642999044}

Rank 4: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.033682295005696056, 'reg__l2_leaf_reg': 15.155331069013117, 'reg__bagging_temperature': 3.497849918015829, 'reg__random_strength': 3.614418483515365, 'reg__subsample': 0.5274079965486366, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 10, 'reg__rsm':

0.8261373683553618, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8736.285653908753}

Rank 5: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.03484918654815463, 'reg__l2_leaf_reg': 10.87166446415656, 'reg__bagging_temperature': 3.3638957248877297, 'reg__random_strength': 3.448177114345759, 'reg__subsample': 0.5019120762860151, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6419012281081787, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8508.305836464451}

Rank 6: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.033986684197502406, 'reg__l2_leaf_reg': 38.06655741156455, 'reg__bagging_temperature': 2.235898121655121, 'reg__random_strength': 2.2444494518517293, 'reg__subsample': 0.5181050469634245, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.6775973303016128, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 952.6443078366658}

Rank 7: {'reg__iterations': 800, 'reg__depth': 8, 'reg__learning_rate': 0.03366678406911247, 'reg__l2_leaf_reg': 7.5662117271598435, 'reg__bagging_temperature': 2.1076037802658996, 'reg__random_strength': 2.5603719155253213, 'reg__subsample': 0.5236831900660525, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6055996870239242, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8380.817542520039}

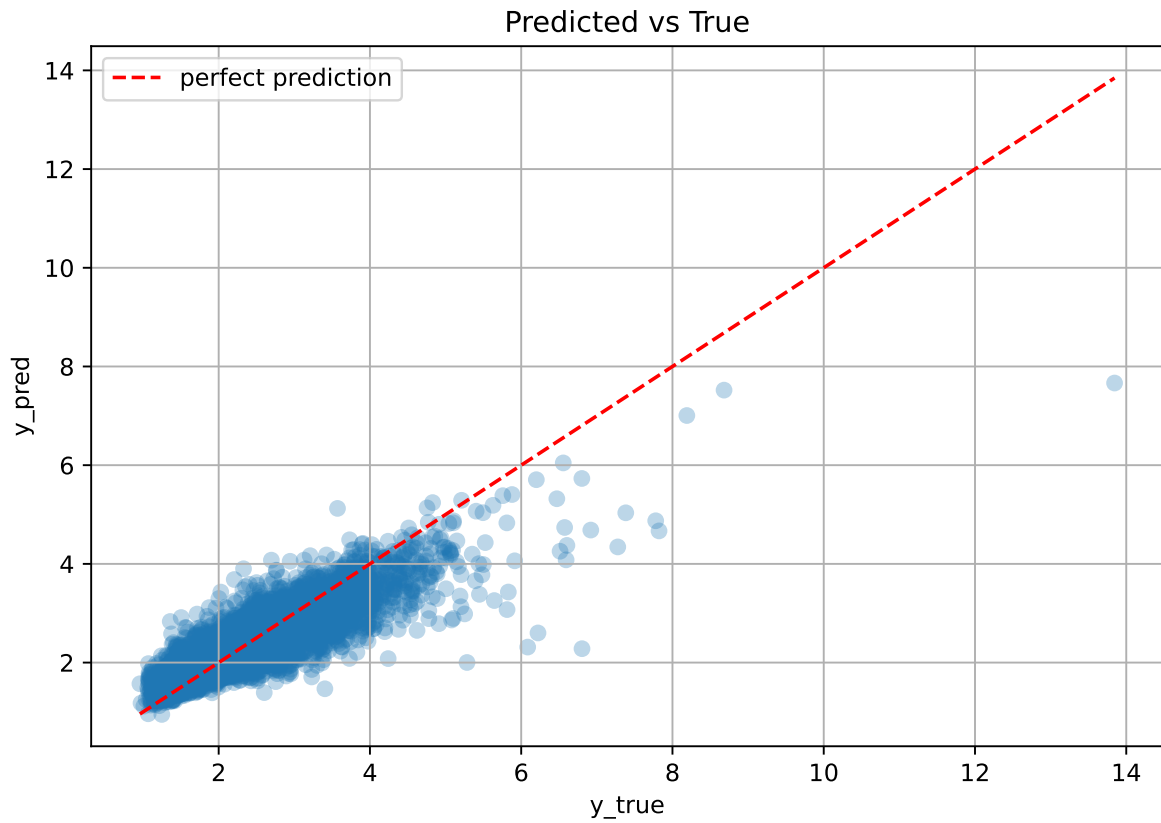
Rank 8: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.02853685996756653, 'reg__l2_leaf_reg': 10.66038260517211, 'reg__bagging_temperature': 2.295983151870793, 'reg__random_strength': 1.9006068446829618, 'reg__subsample': 0.5003742557067138, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6830014622435117, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7830.62228430322}

Rank 9: {'reg__iterations': 800, 'reg__depth': 8, 'reg__learning_rate': 0.029311814634369503, 'reg__l2_leaf_reg': 11.500753780122189, 'reg__bagging_temperature': 4.238590087968751, 'reg__random_strength': 3.381340557189195, 'reg__subsample': 0.5518922648458097, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6131269540053206, 'reg__border_count': 125, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7687.873683687633}

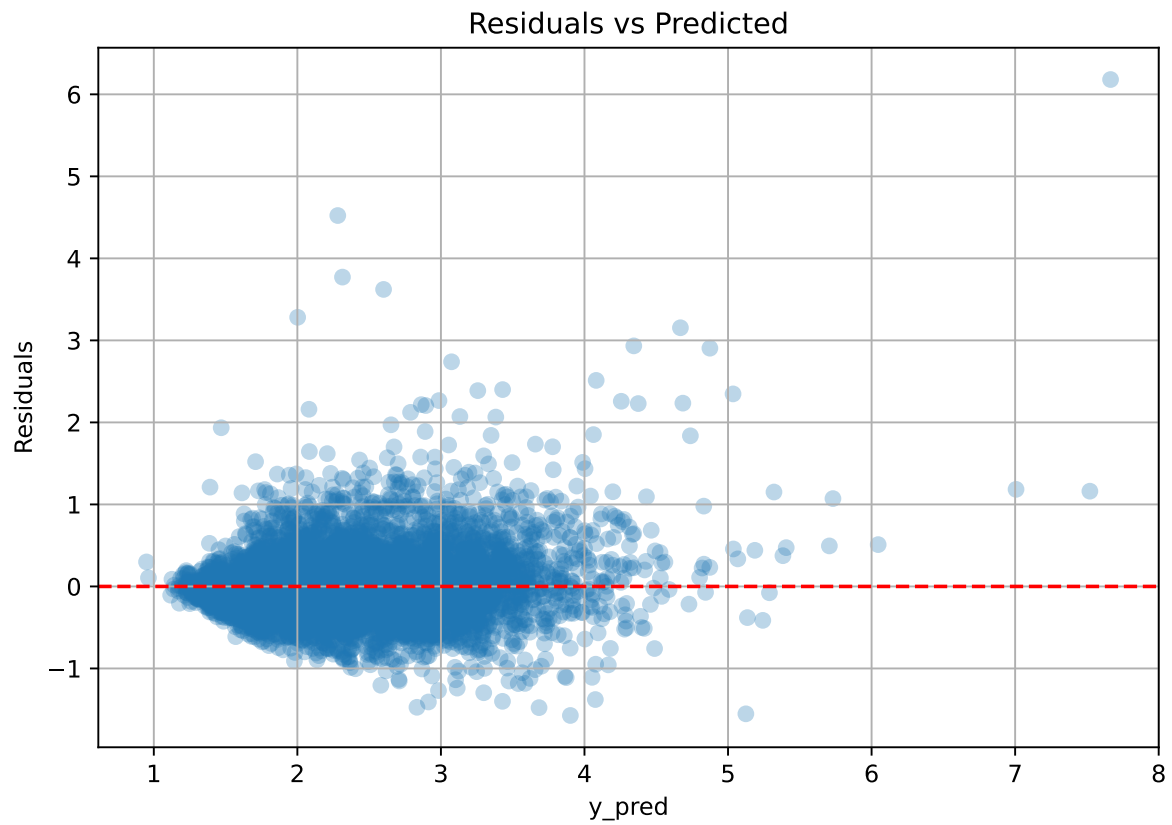
Rank 10: {'reg__iterations': 900, 'reg__depth': 8, 'reg__learning_rate': 0.03007824266670564, 'reg__l2_leaf_reg': 10.183858724847285, 'reg__bagging_temperature': 3.40420450439491,

```
'reg__random_strength': 1.9697998968793624, 'reg__subsample': 0.5016781443151284,  
'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 9, 'reg__rsm':  
0.6427181611461779, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature':  
7960.391616030223}
```

2.5.2 Scatter



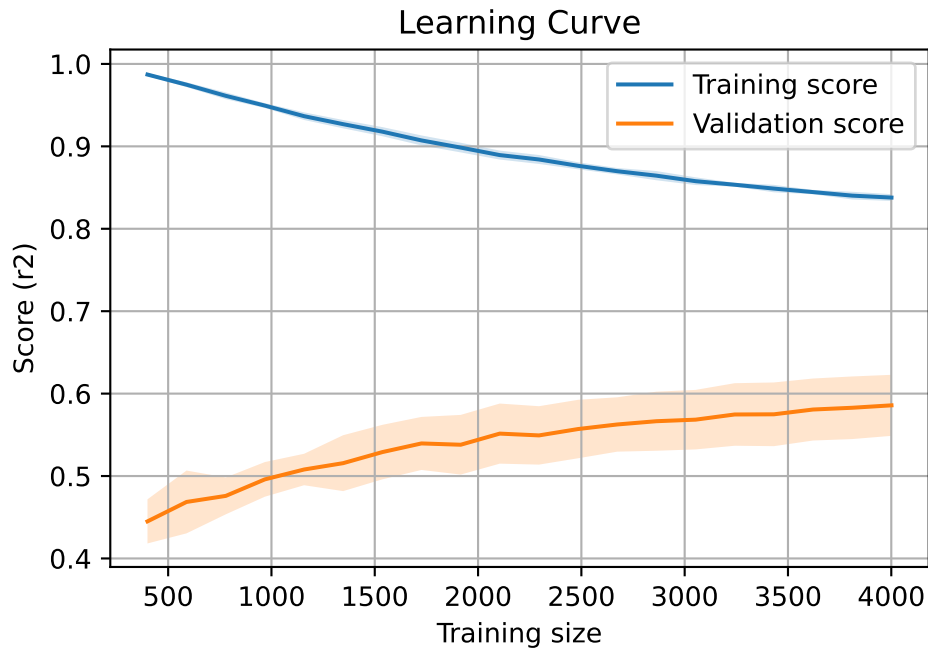
2.5.3 Scatter residuals



2.5.4 Learning curve

`Len(group)` : 8168, `Len(training)` : 5718,

`Len(training_real)` : 4003, `Len(test_of_training)` : 1715, `Len(test)` : 2450



2.6 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.36	0.365	0.037	0.345	0.345	-0.02	0.946	0.986	nan
LGBM	0.697	0.564	0.039	0.599	0.612	0.049	1.087	1.238	0.5635
CatBoost	0.836	0.586	0.04	0.624	0.638	0.052	1.09	1.428	0.5858

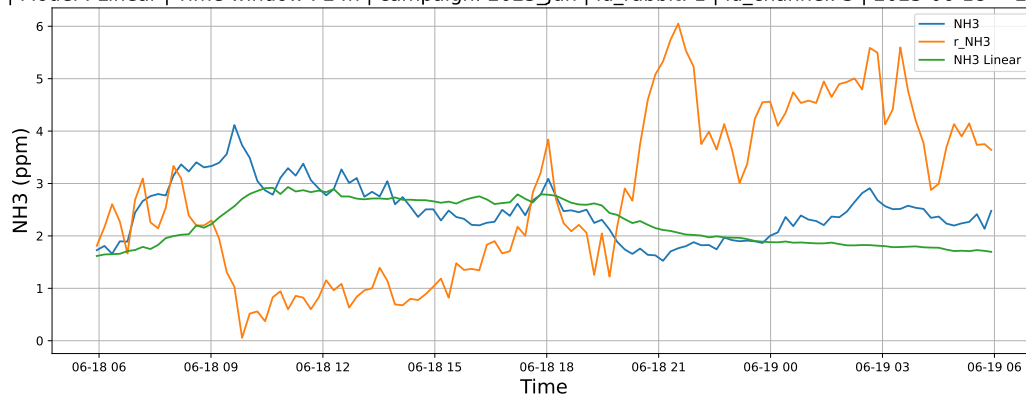
3 Plot

3.1 Standalone plots

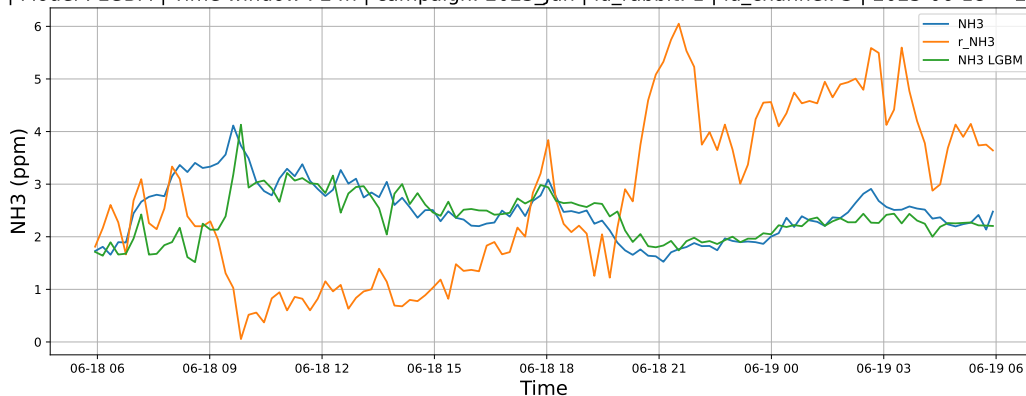
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

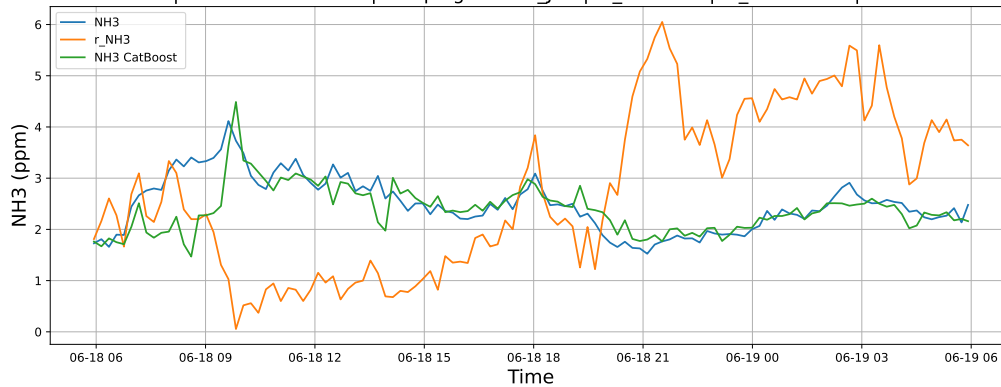
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

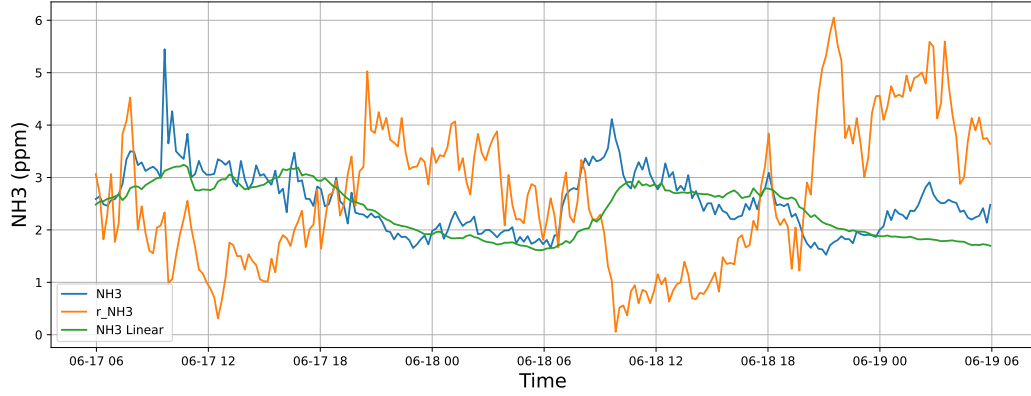


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

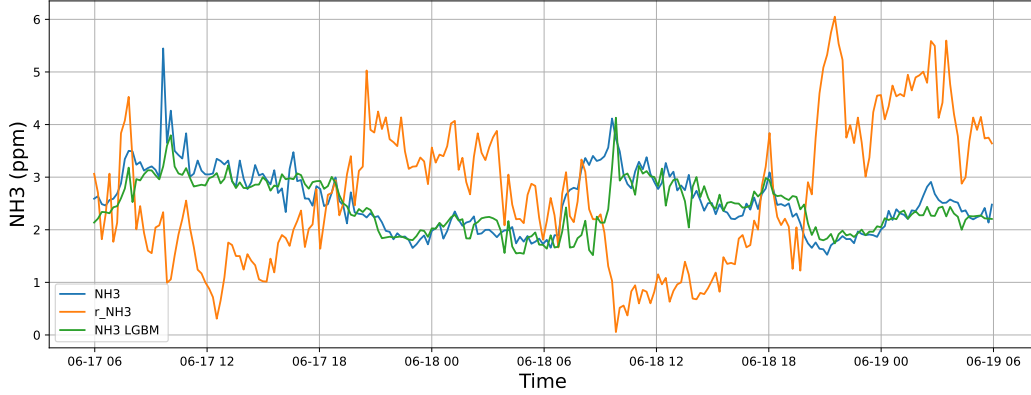


3.1.1.2 Time window : 48

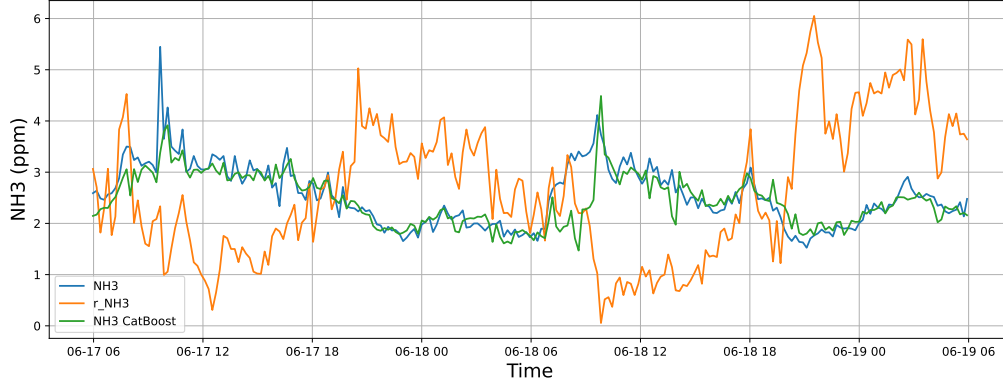
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

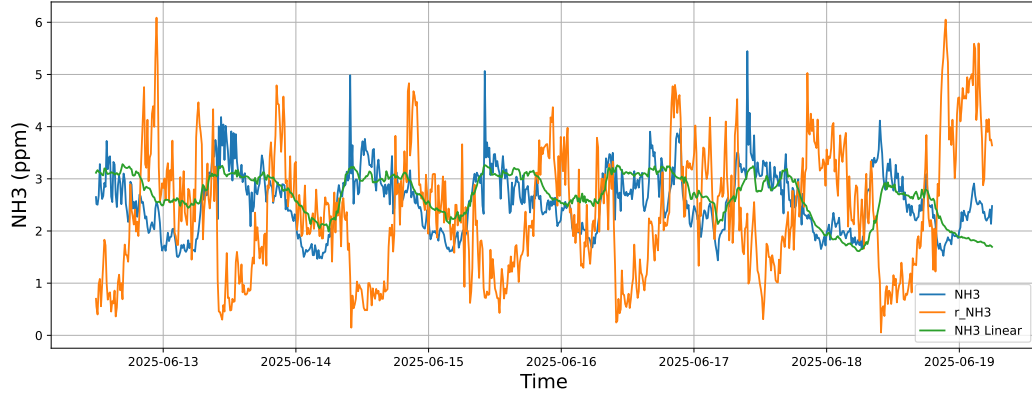


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

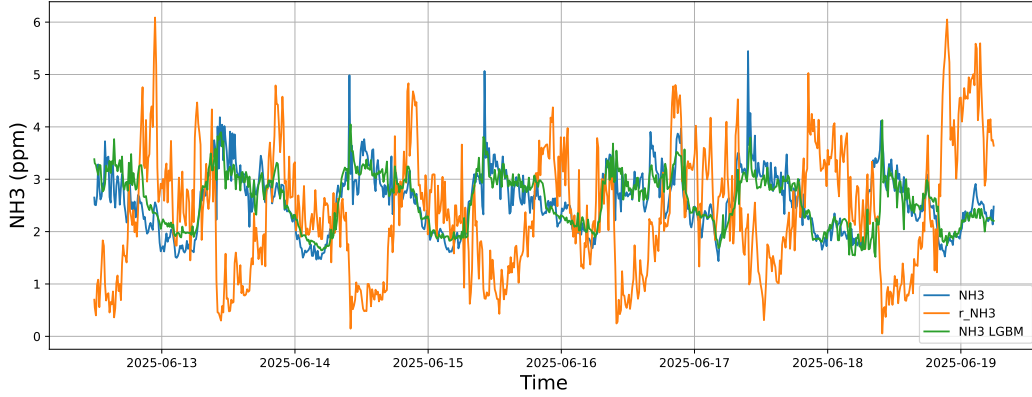


3.1.1.3 Time window : ALL

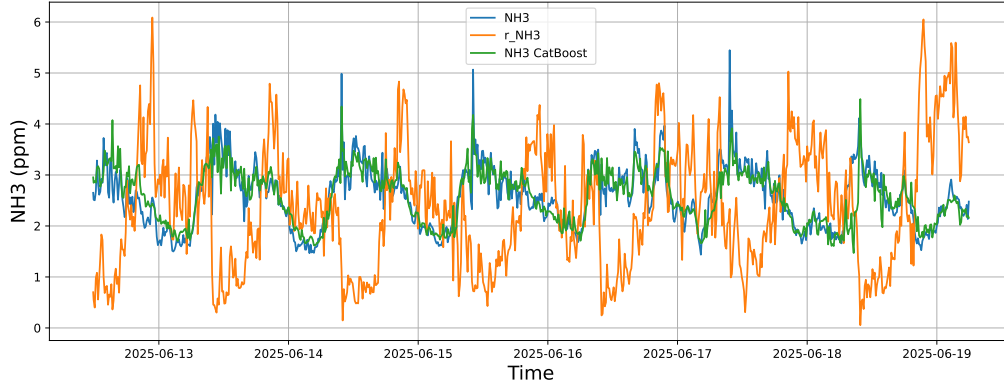
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



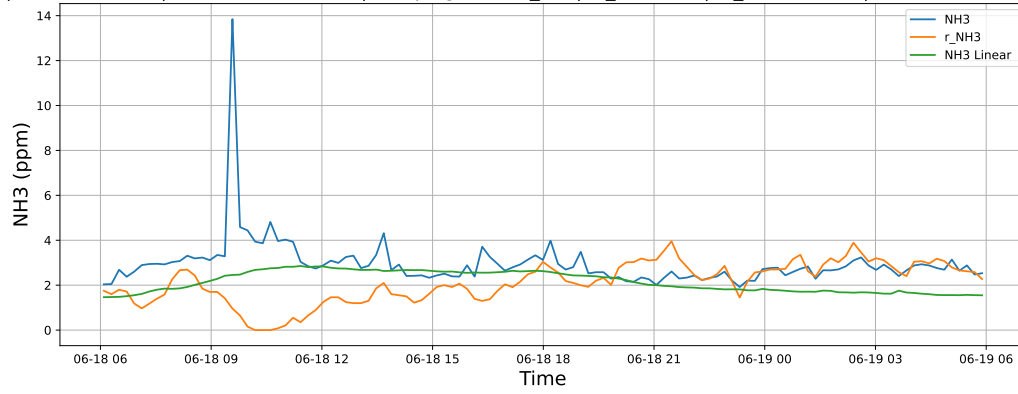
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



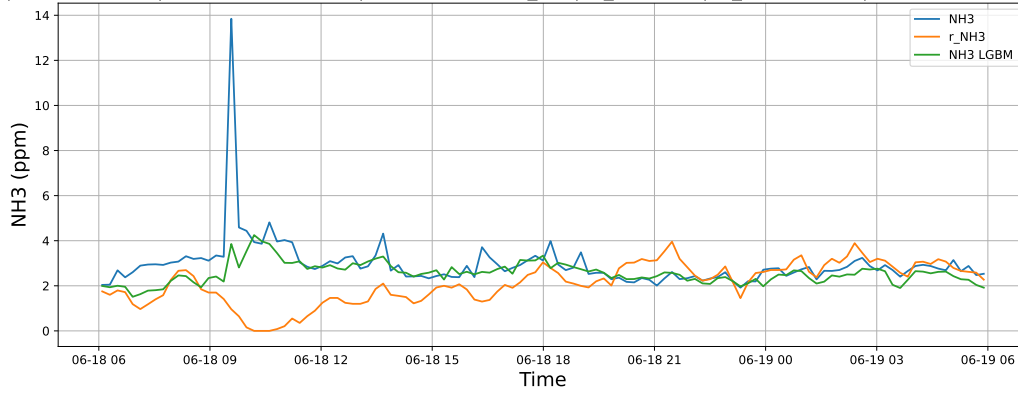
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

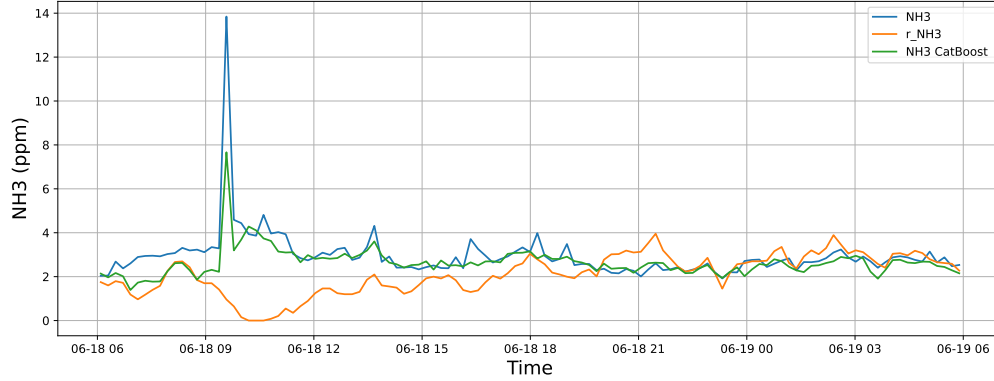
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

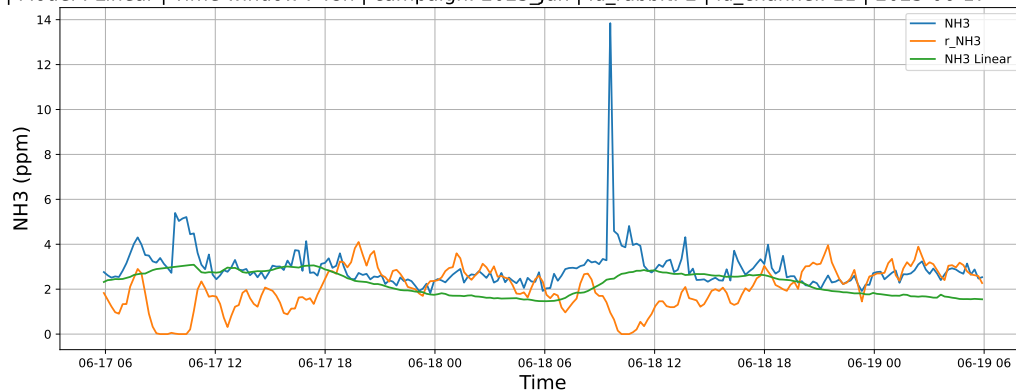


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

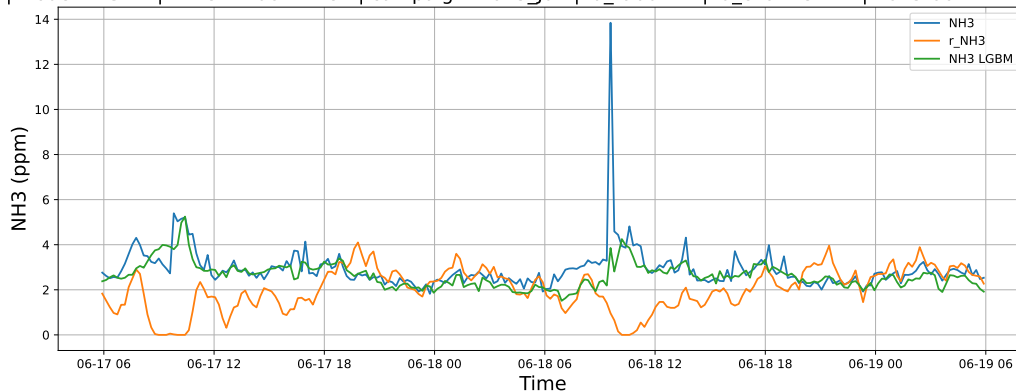


3.1.2.2 Time window : 48

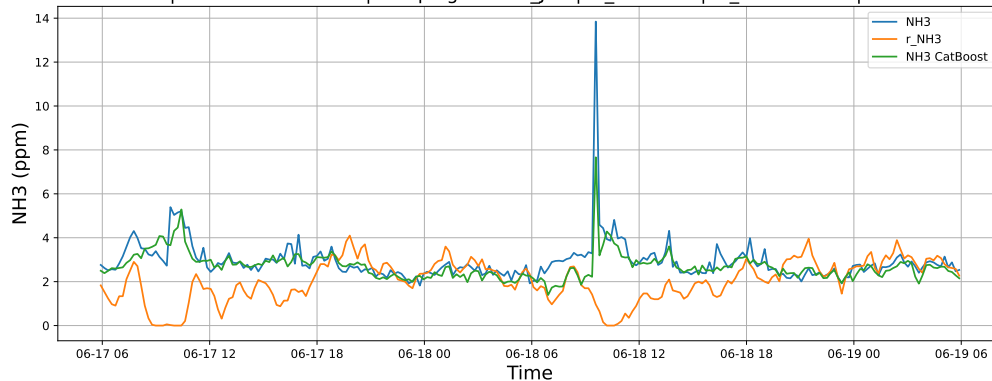
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

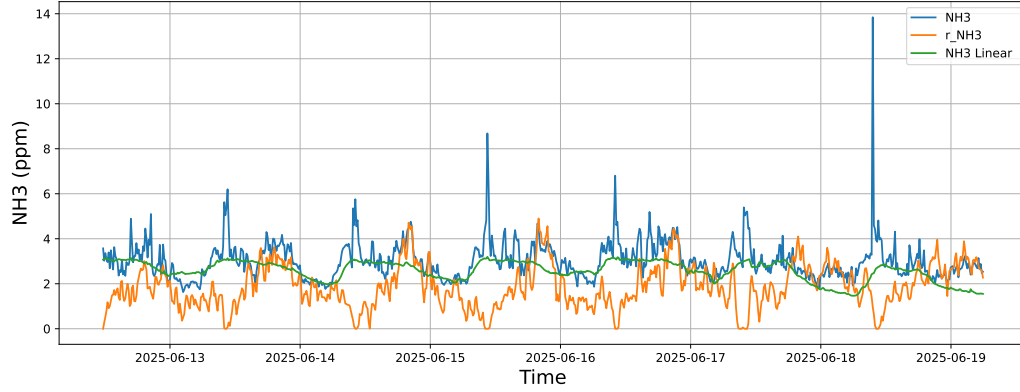


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

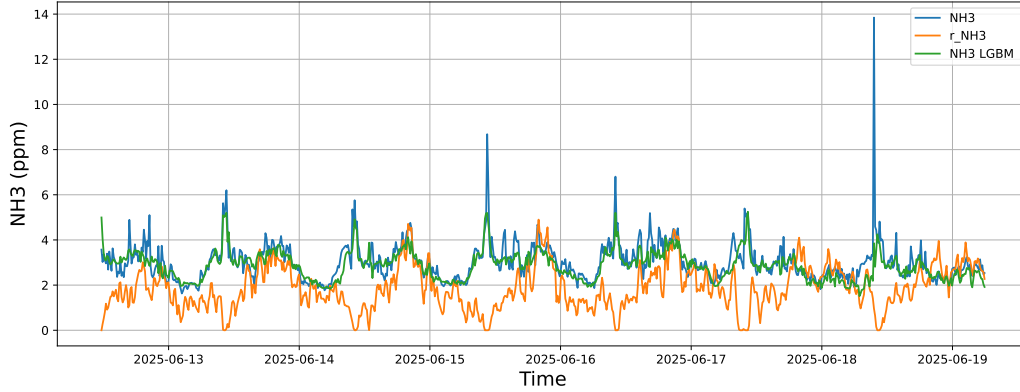


3.1.2.3 Time window : ALL

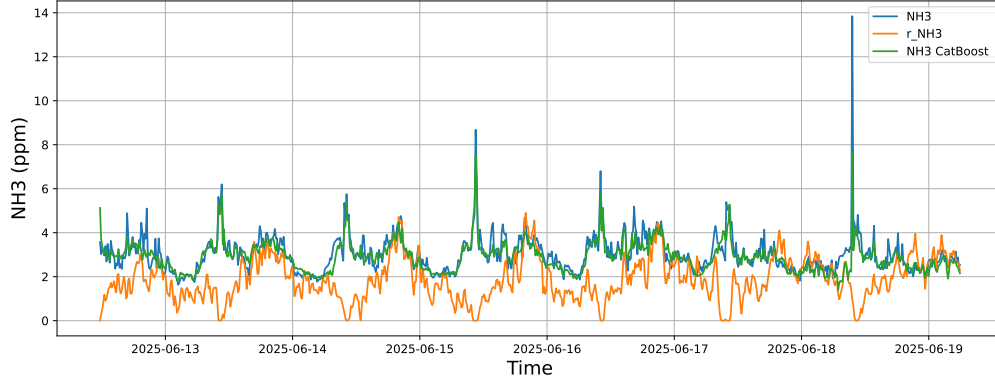
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



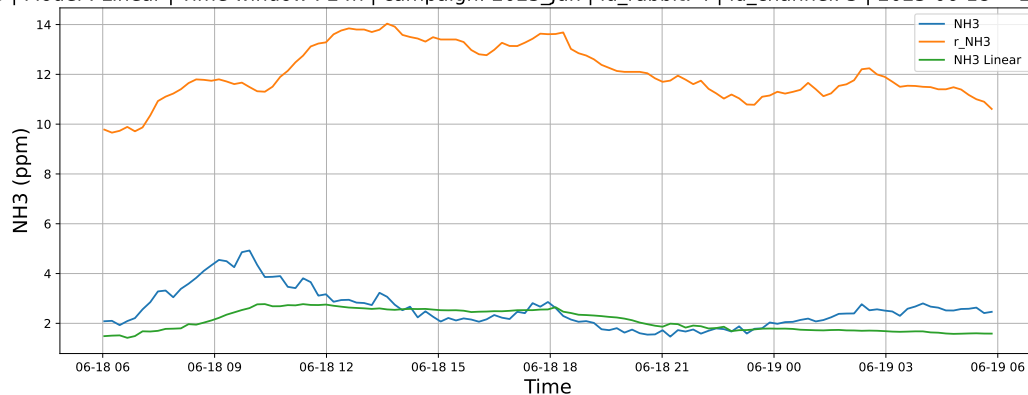
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



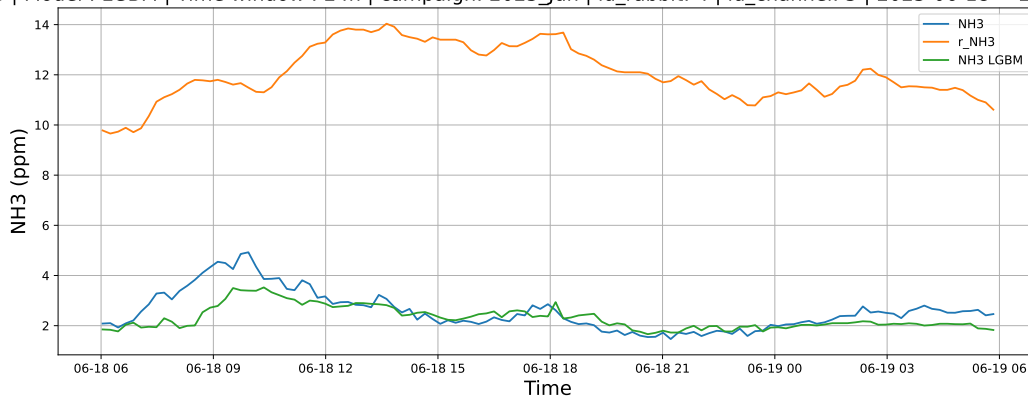
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

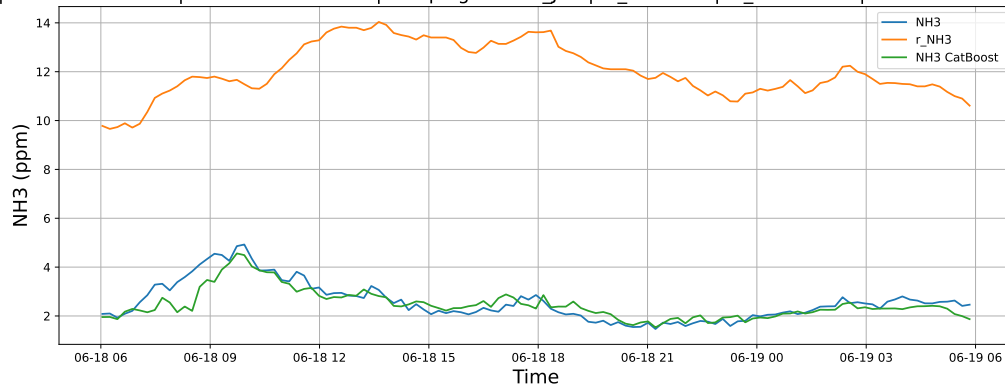
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

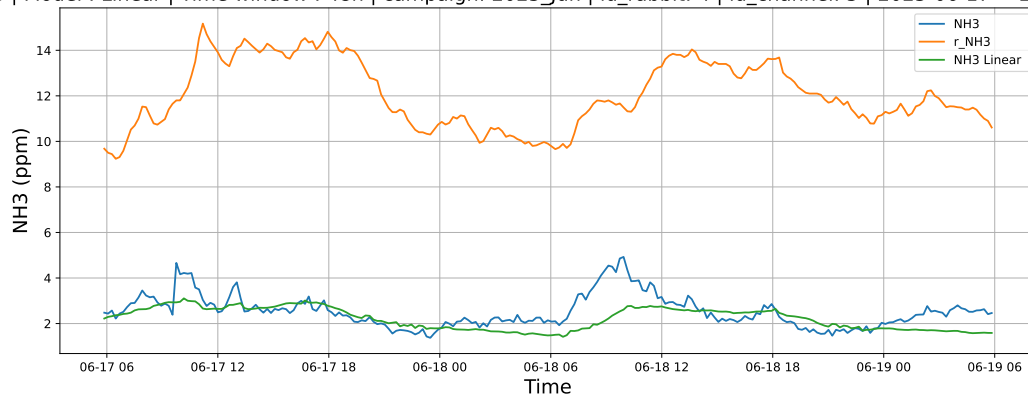


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

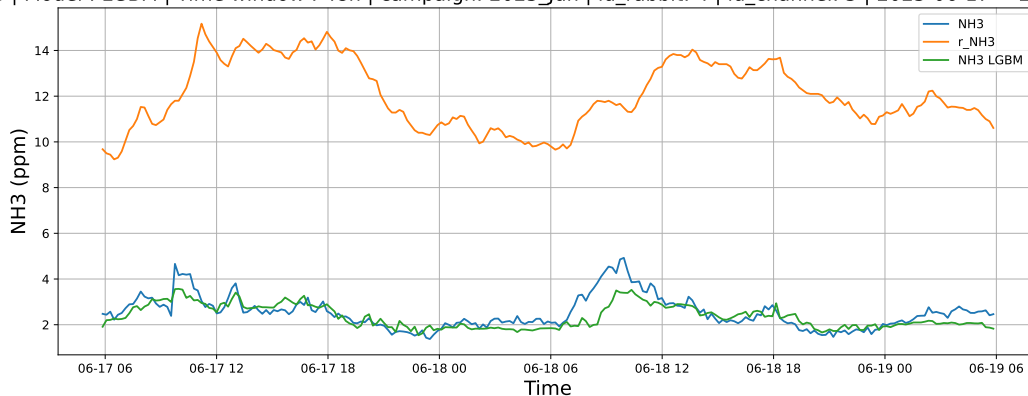


3.1.3.2 Time window : 48

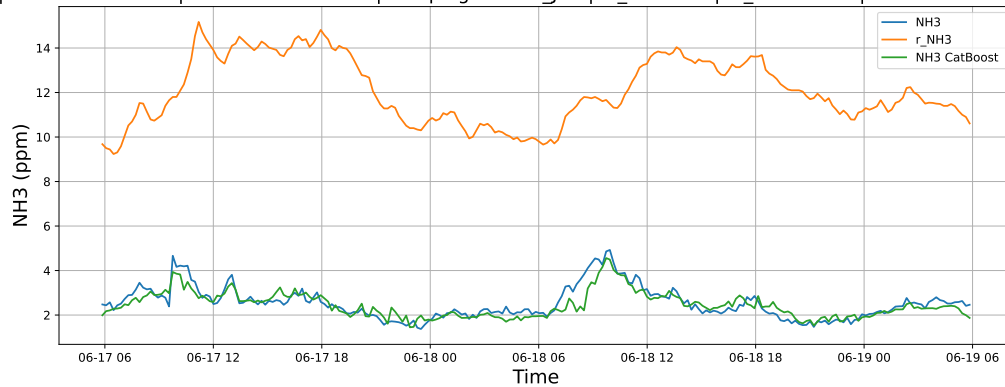
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

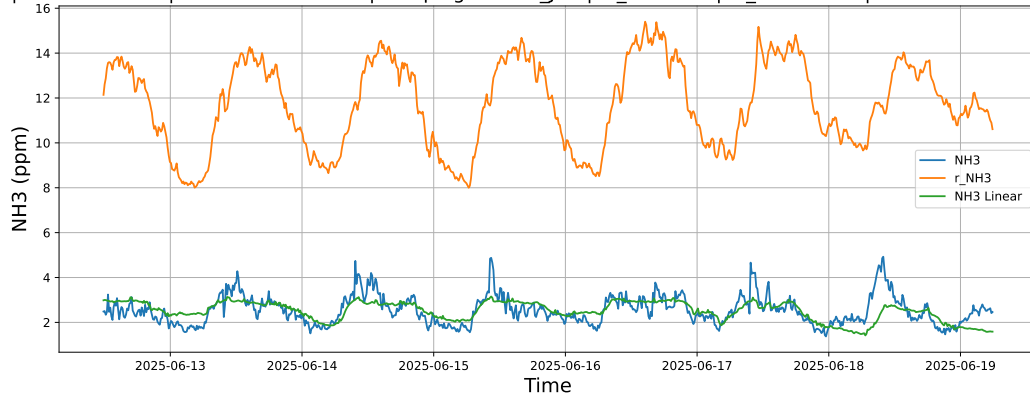


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

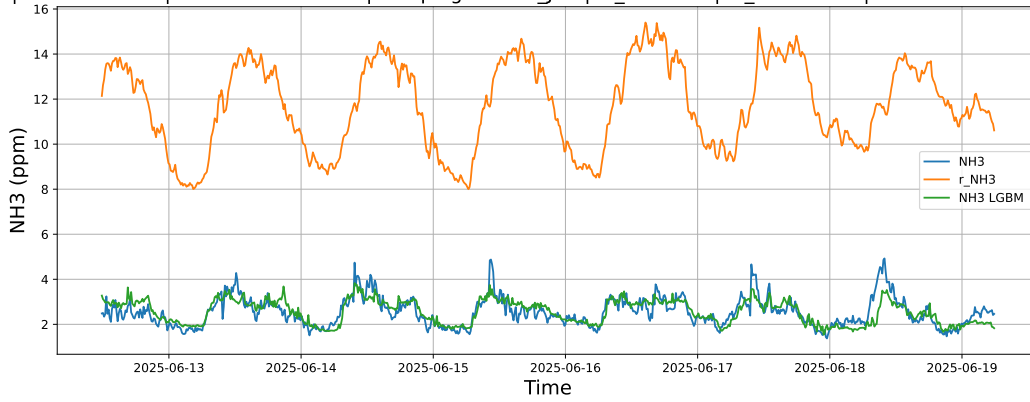


3.1.3.3 Time window : ALL

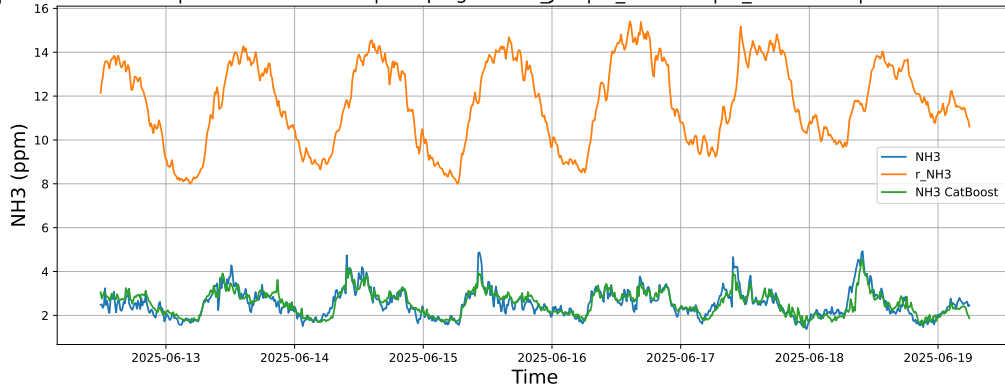
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



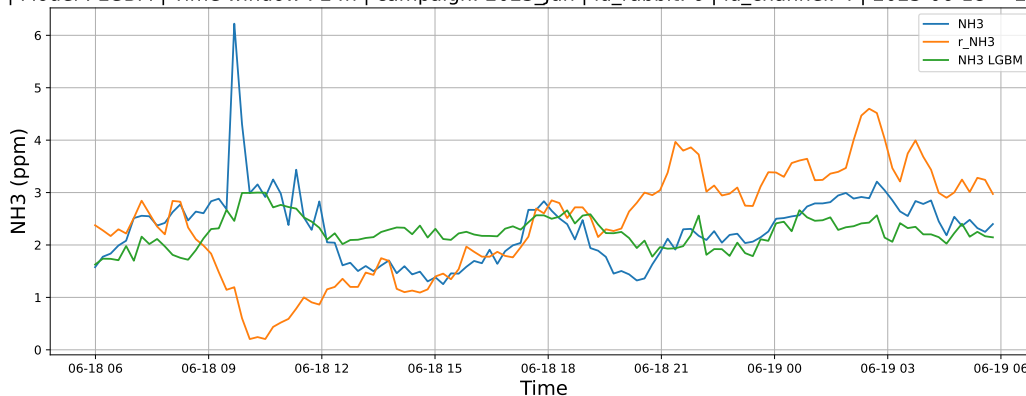
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

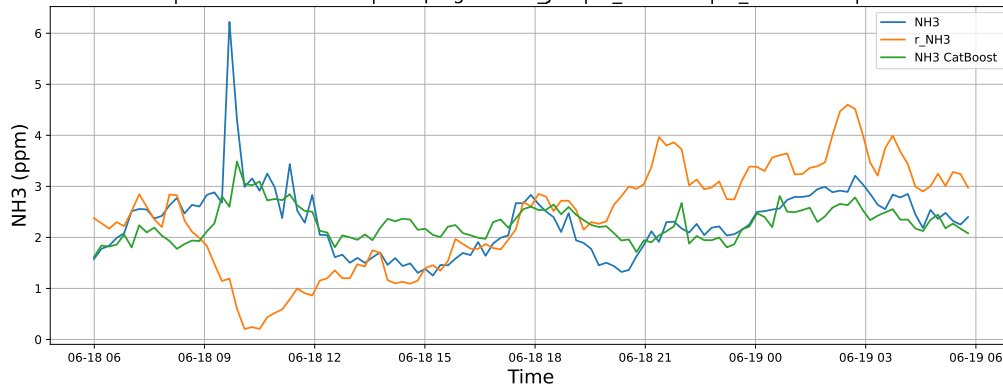
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

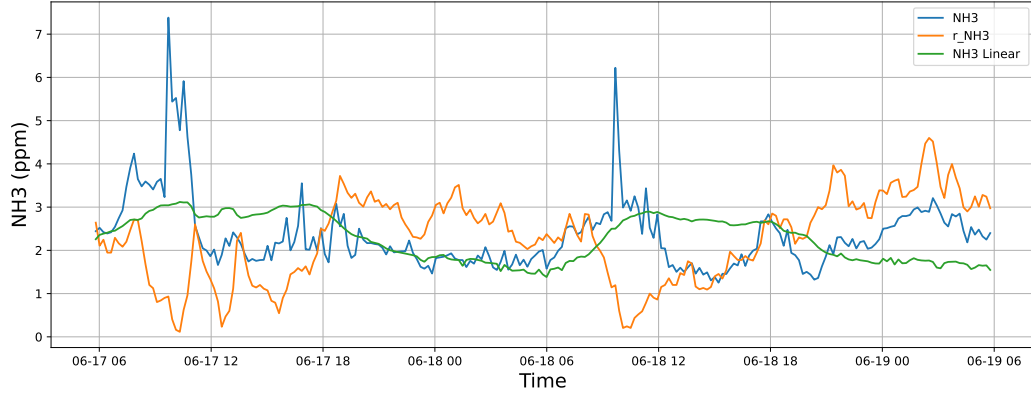


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

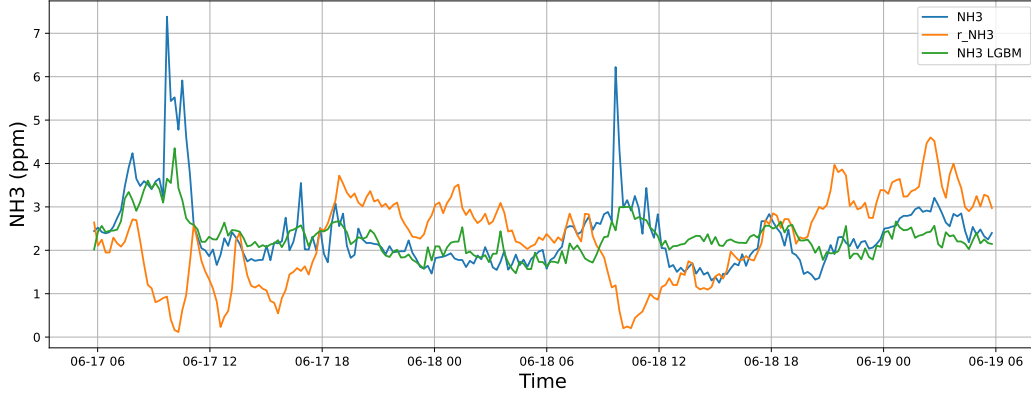


3.1.4.2 Time window : 48

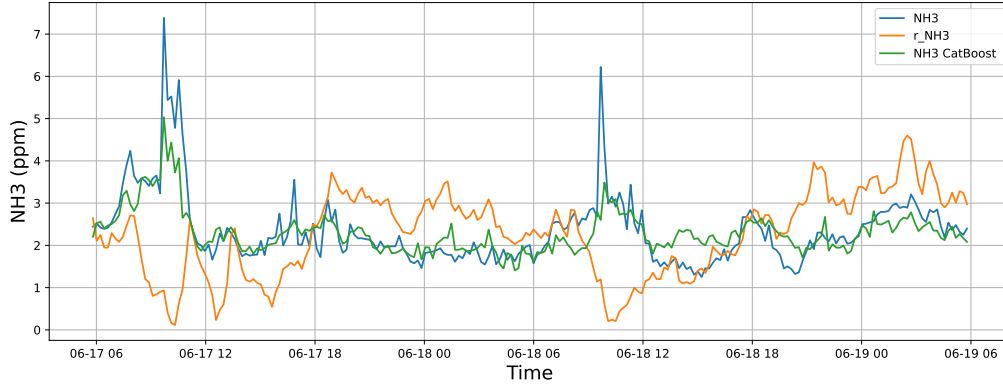
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

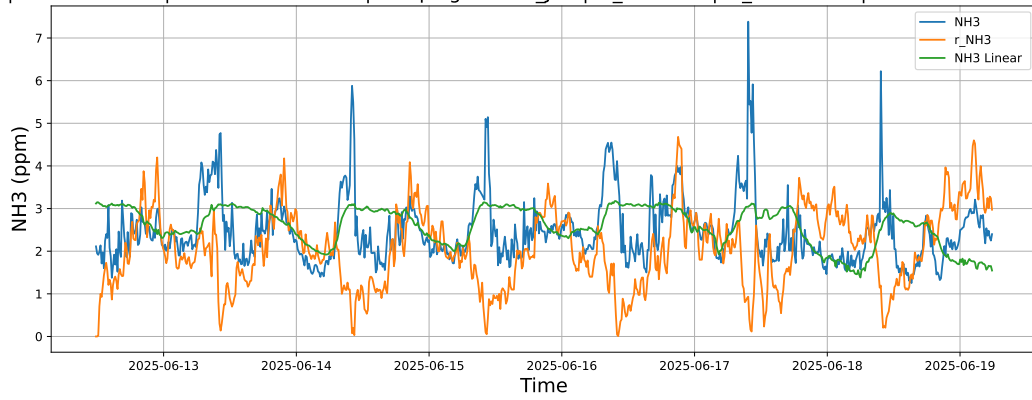


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

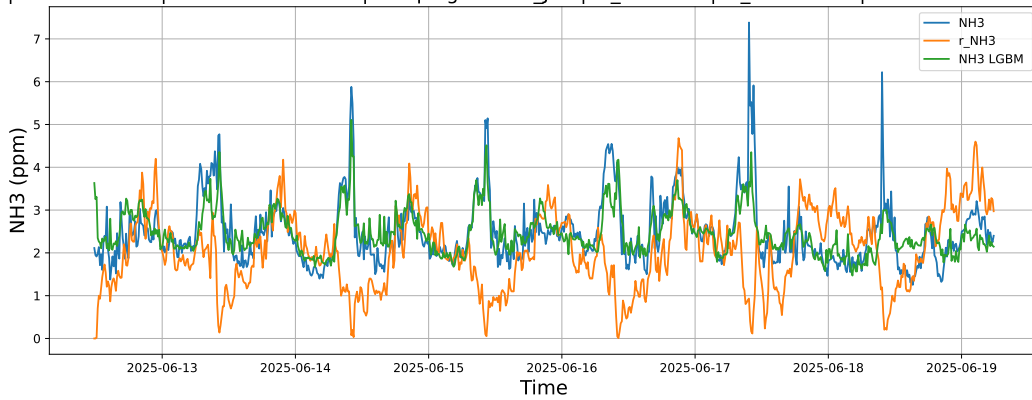


3.1.4.3 Time window : ALL

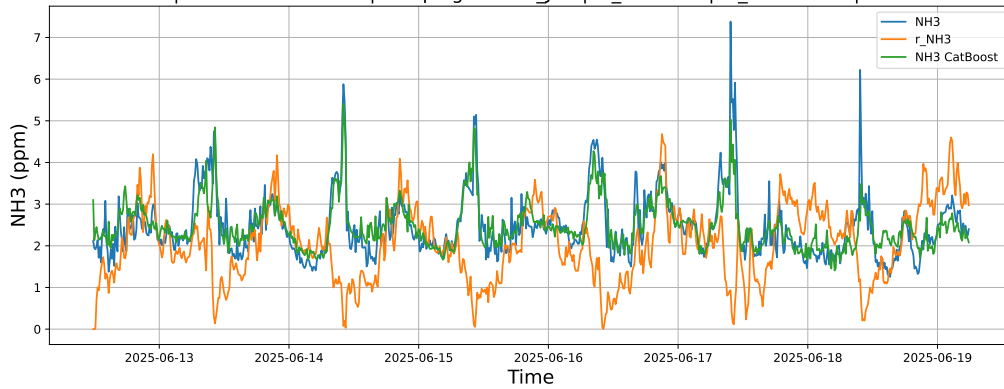
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



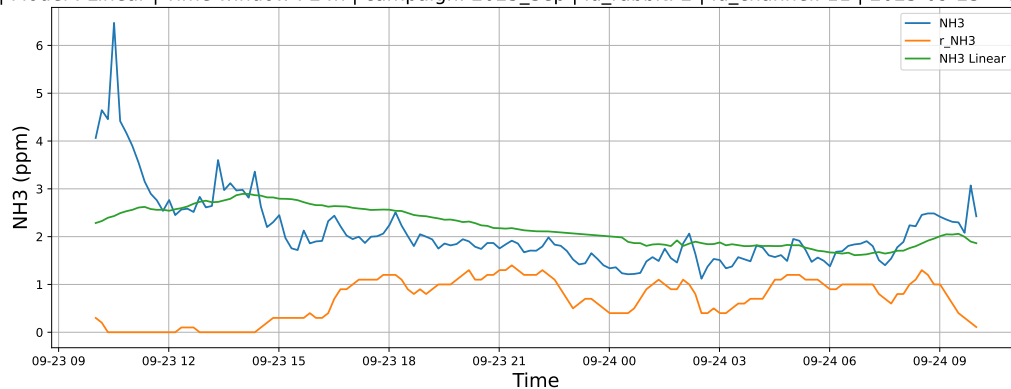
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



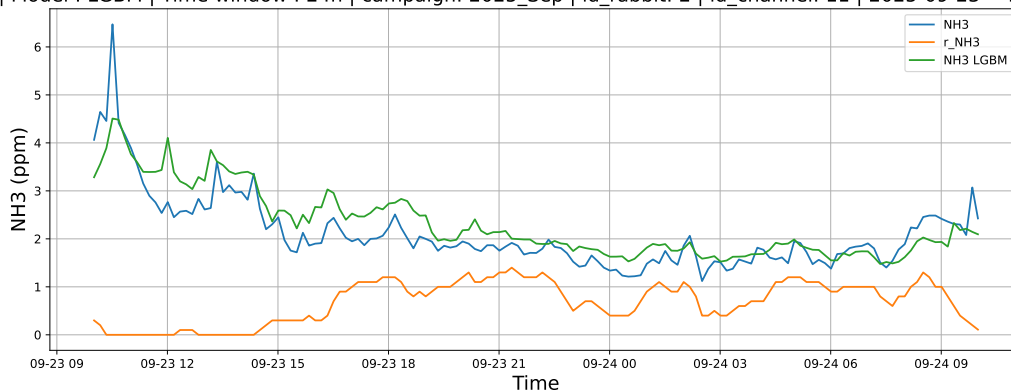
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

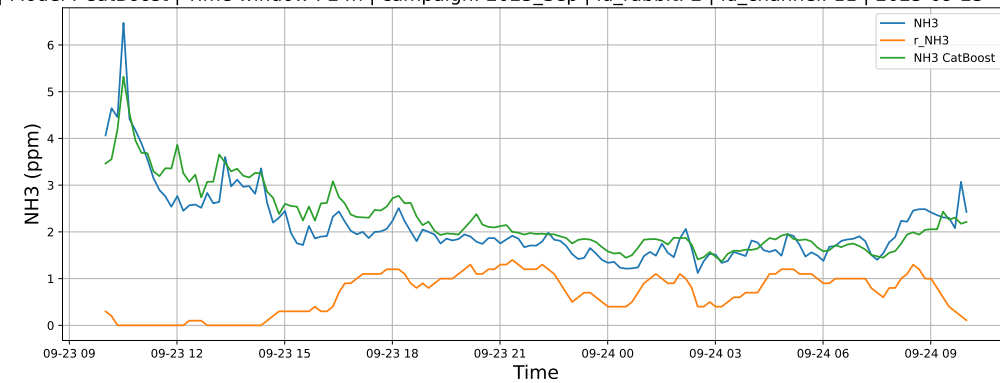
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

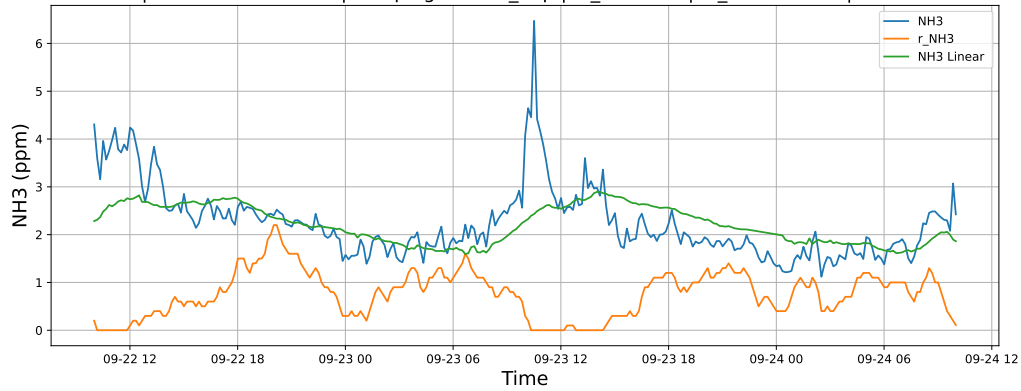


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

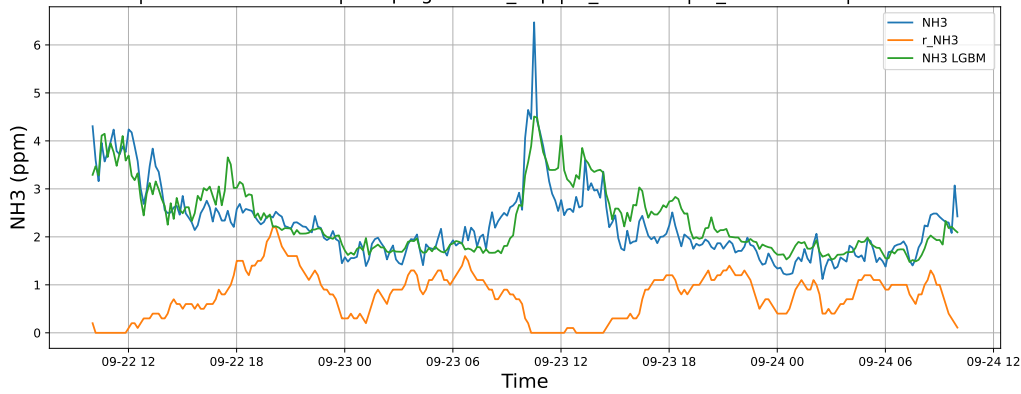


3.1.5.2 Time window : 48

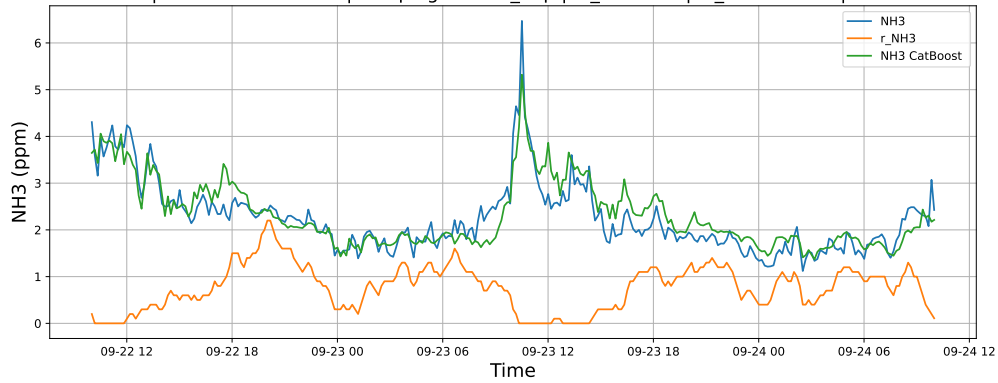
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

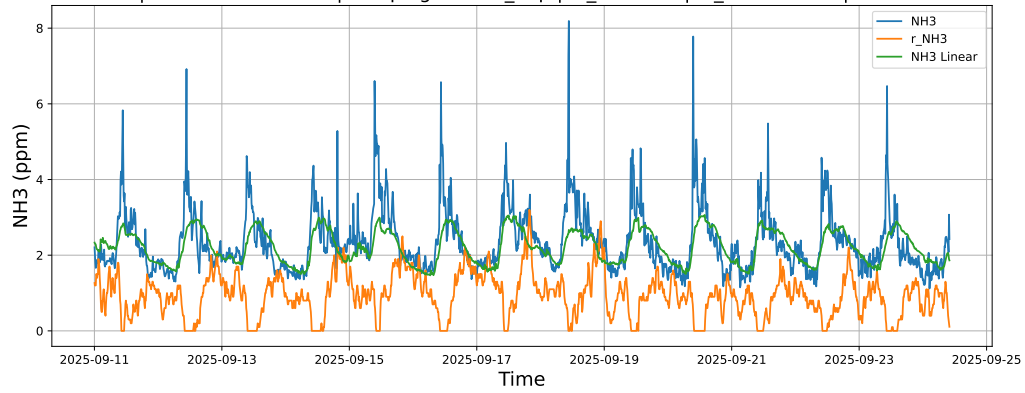


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

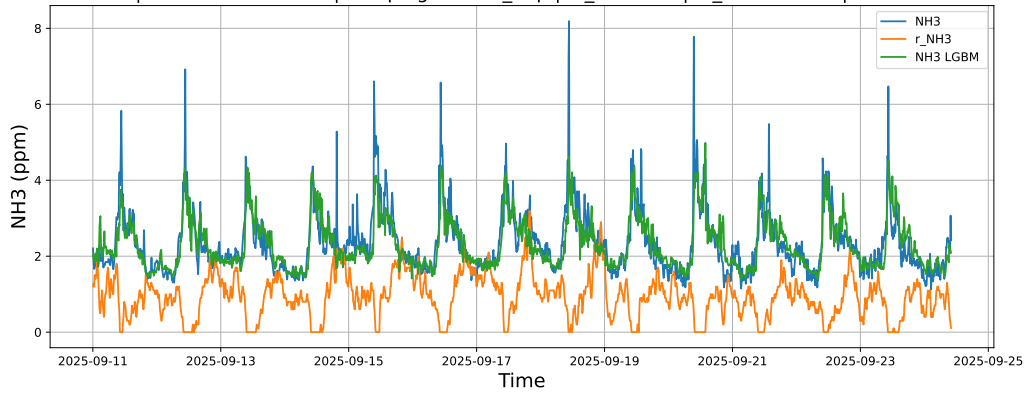


3.1.5.3 Time window : ALL

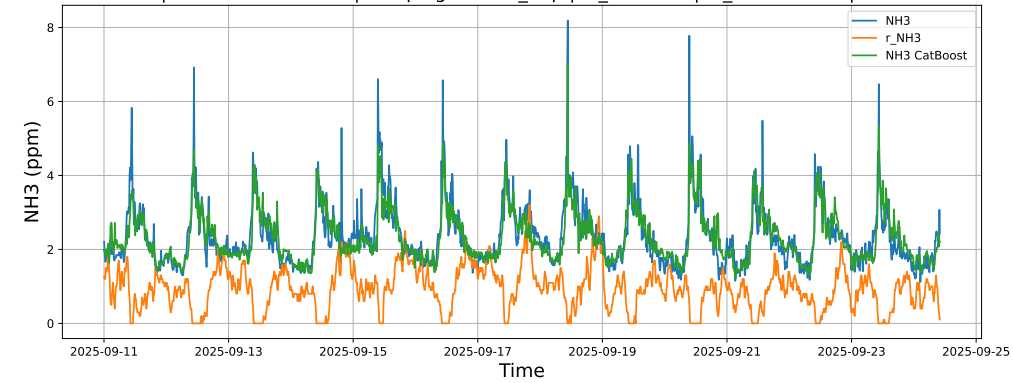
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



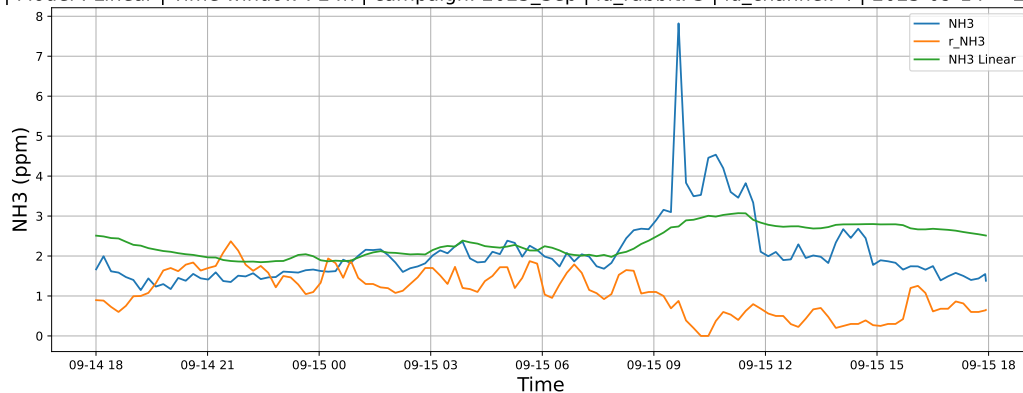
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



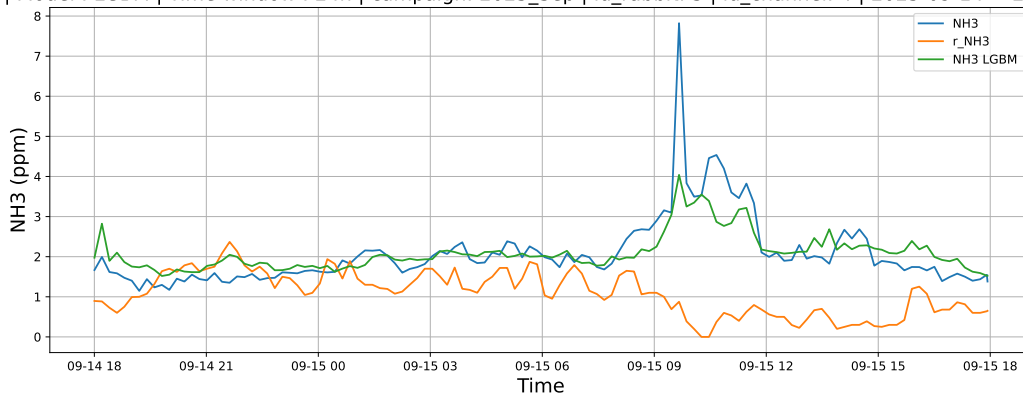
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

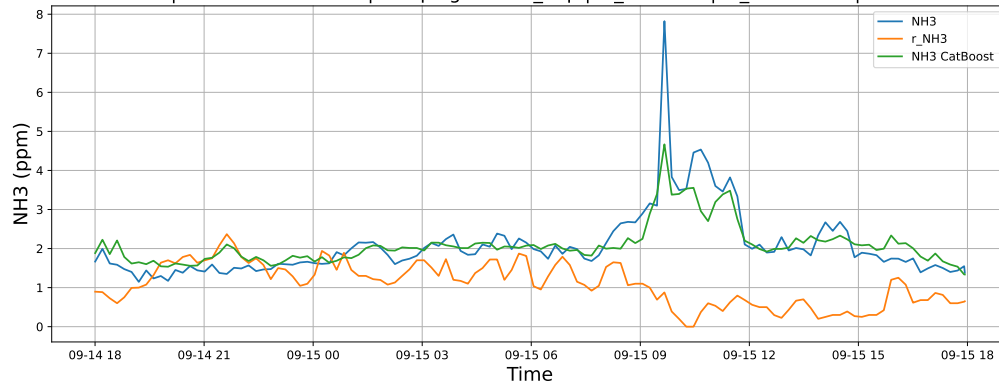
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

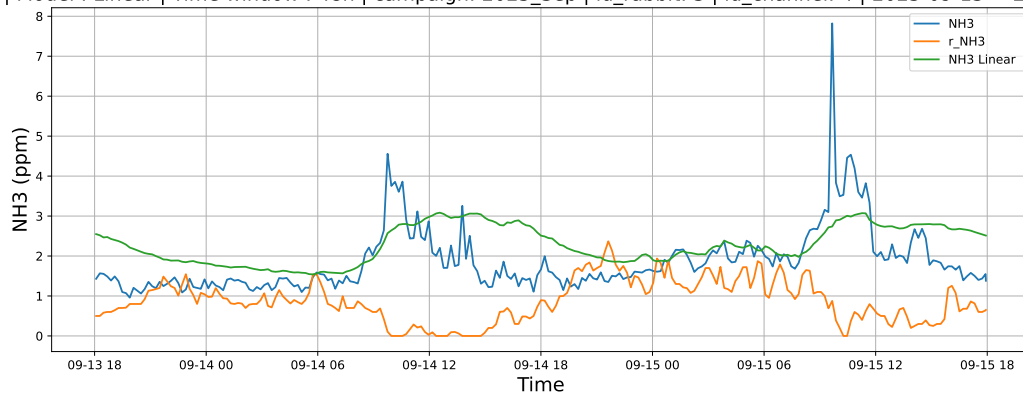


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

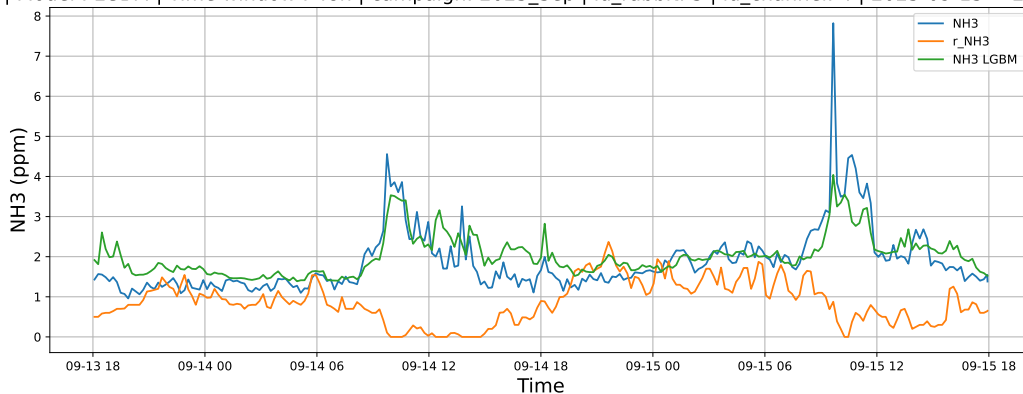


3.1.6.2 Time window : 48

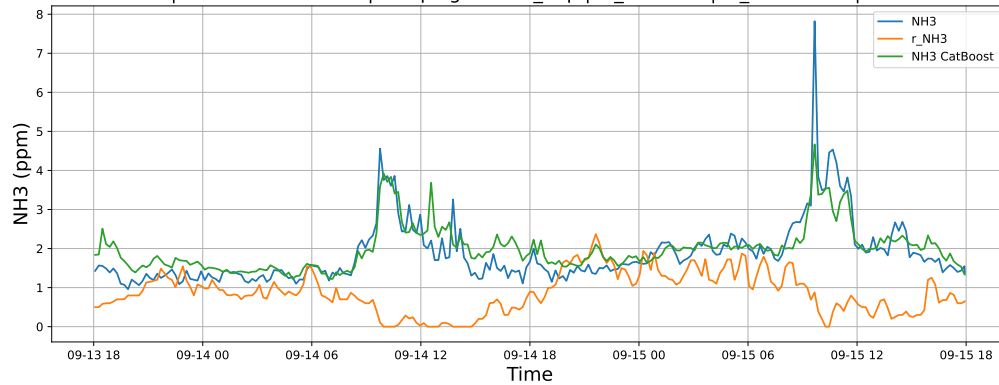
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

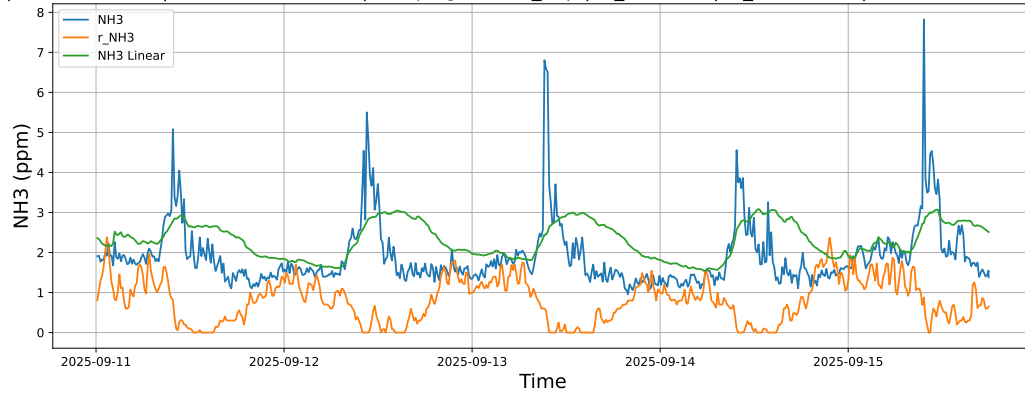


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

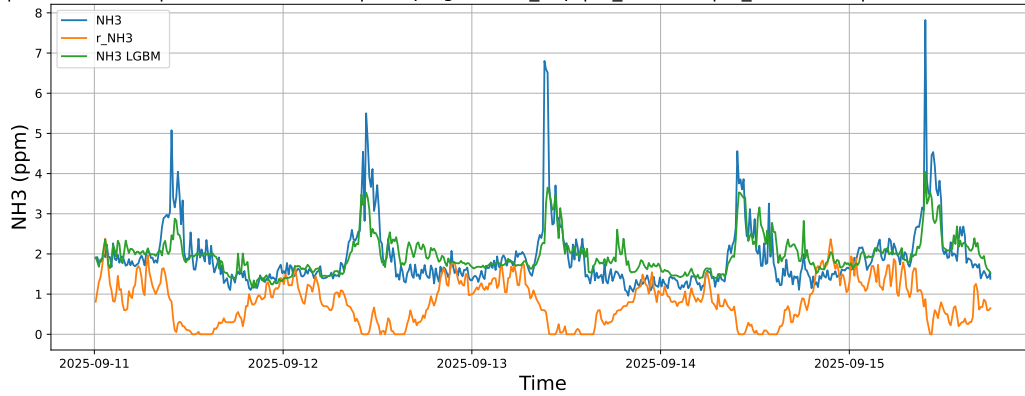


3.1.6.3 Time window : ALL

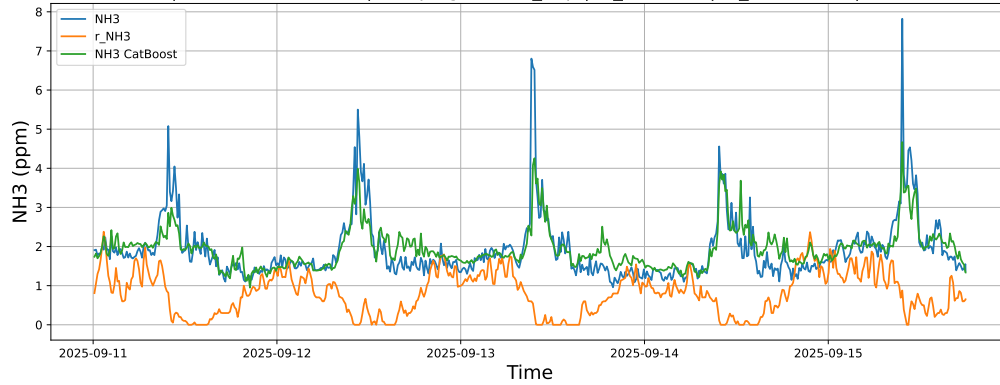
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



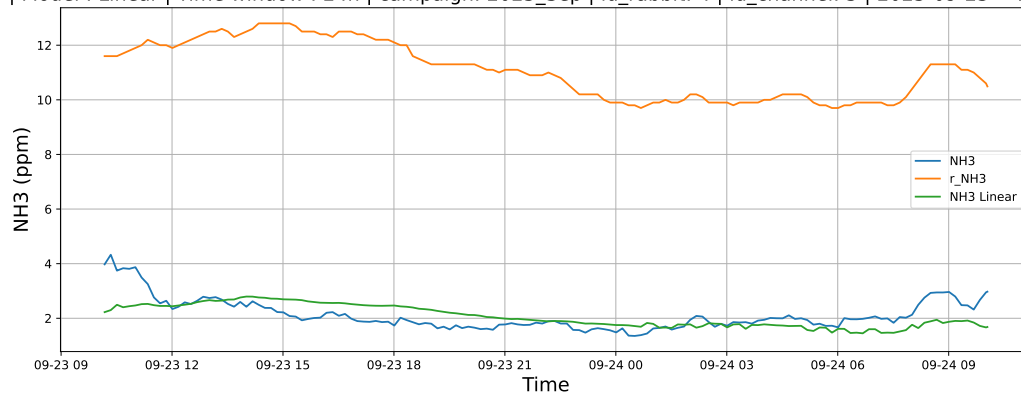
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



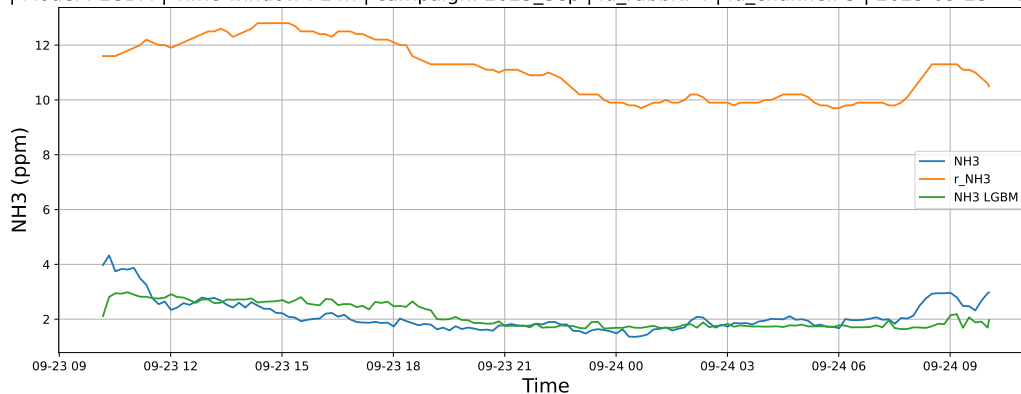
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

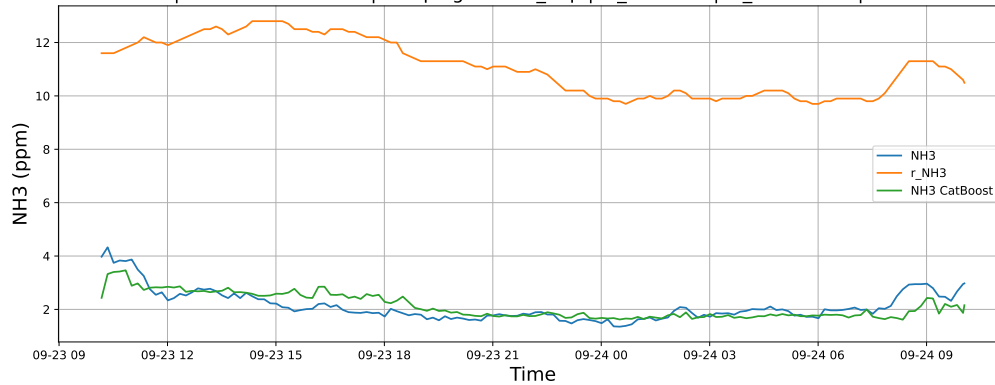
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

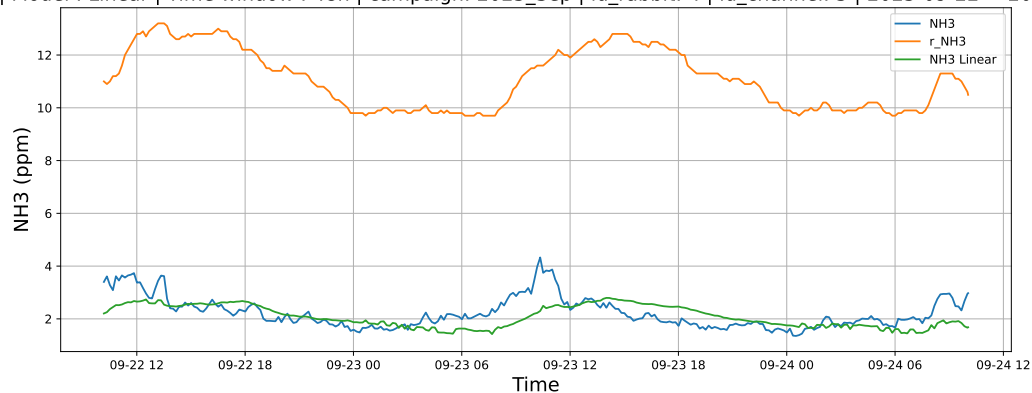


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

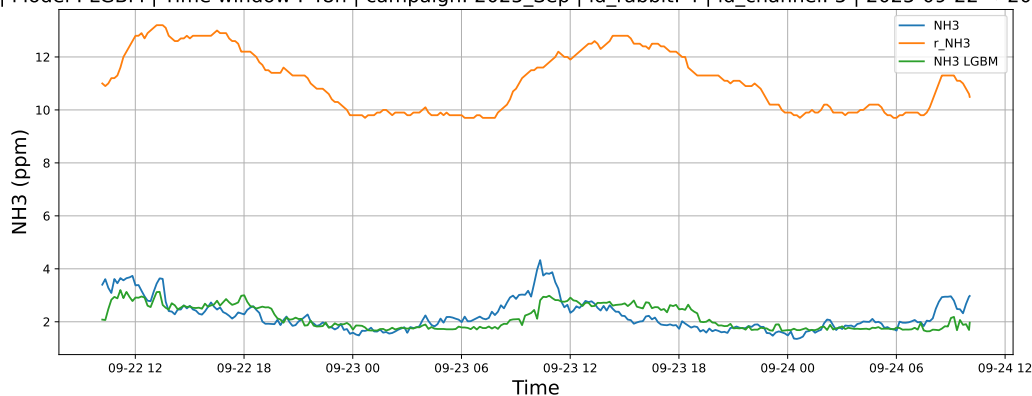


3.1.7.2 Time window : 48

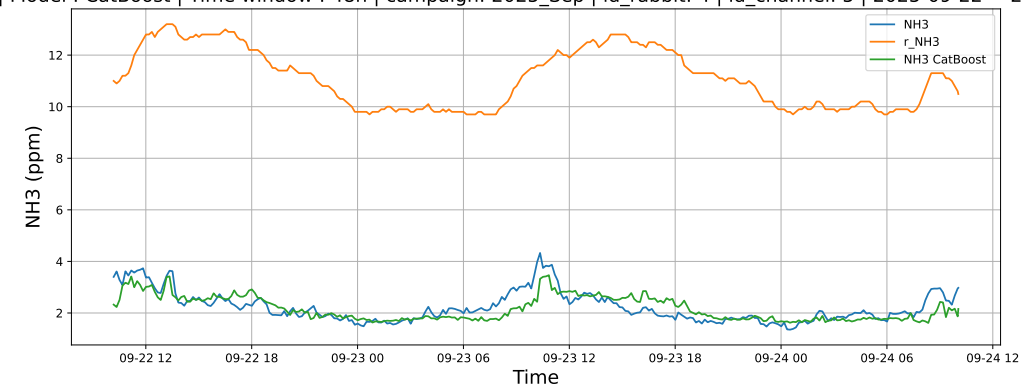
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

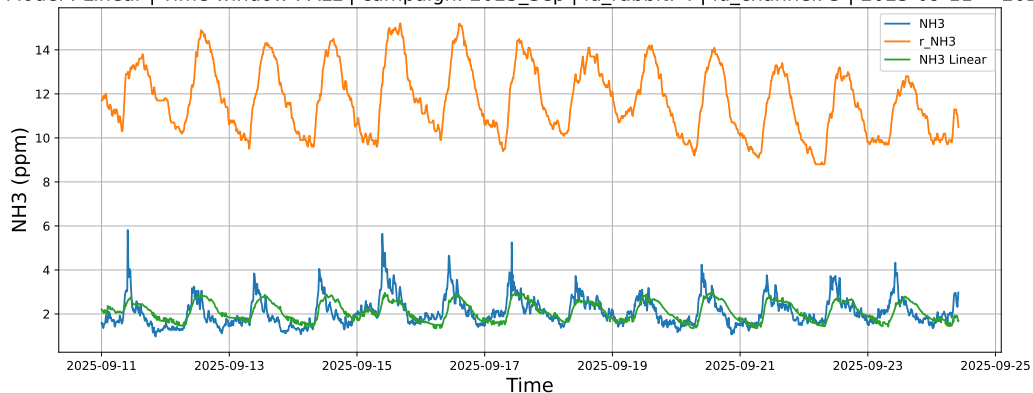


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

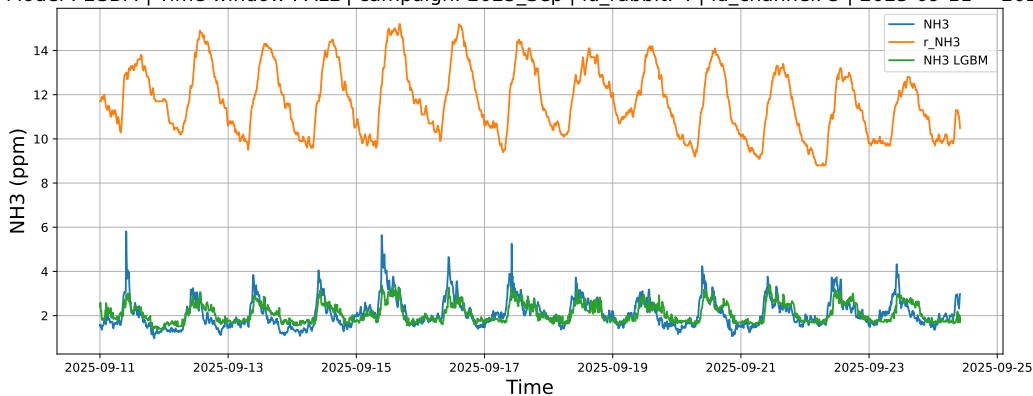


3.1.7.3 Time window : ALL

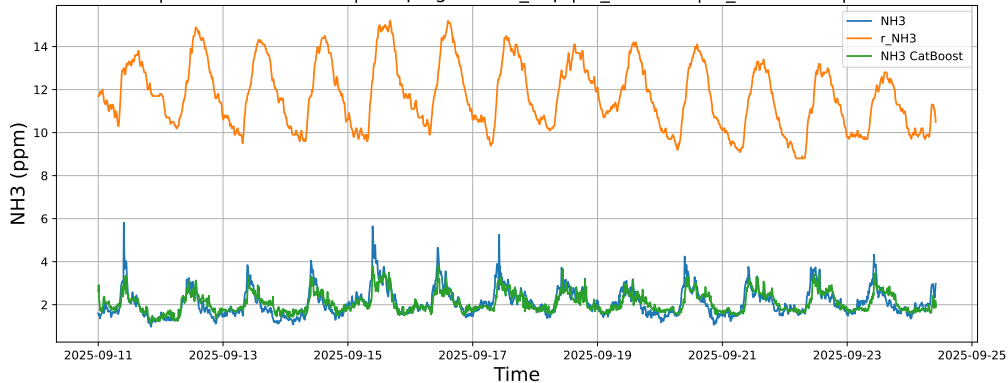
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



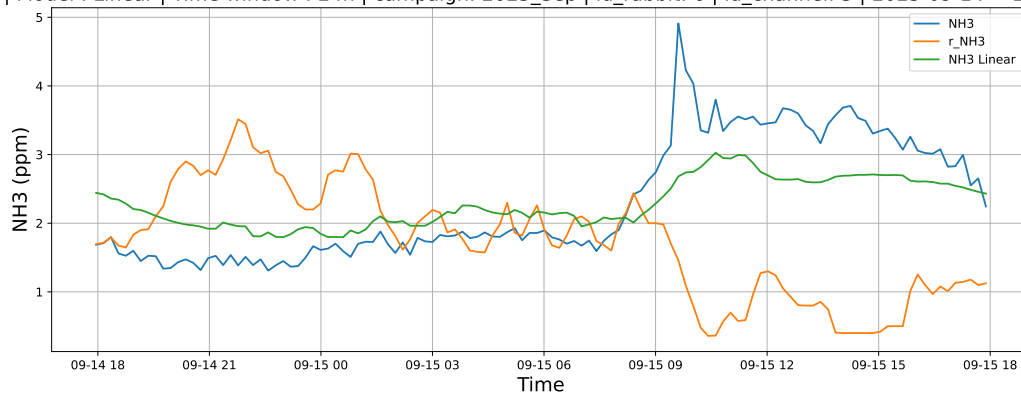
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



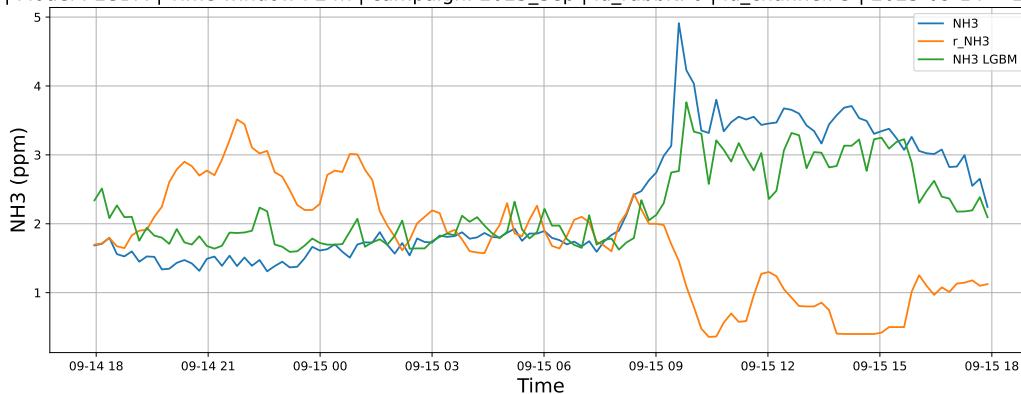
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

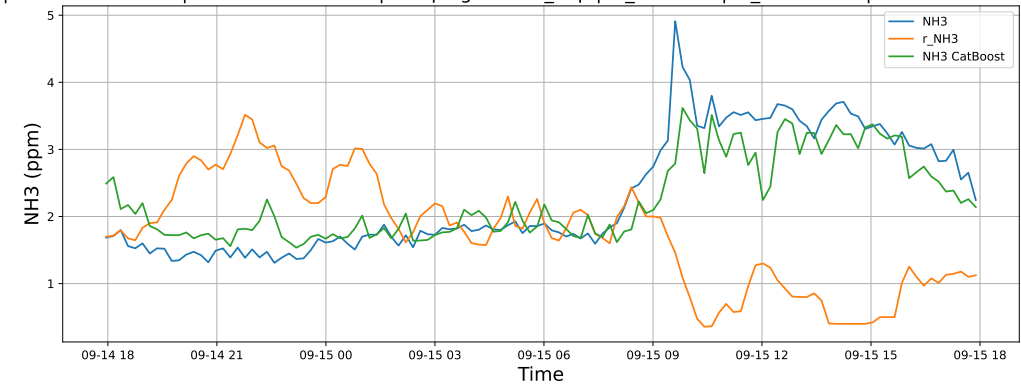
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

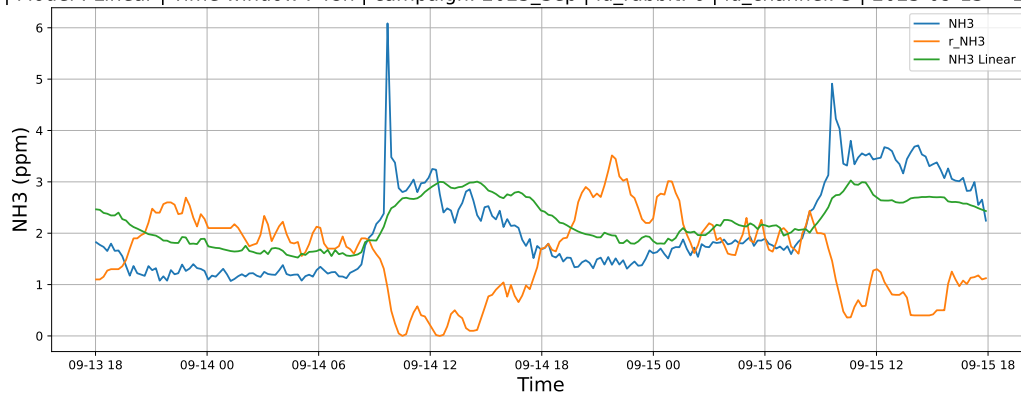


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

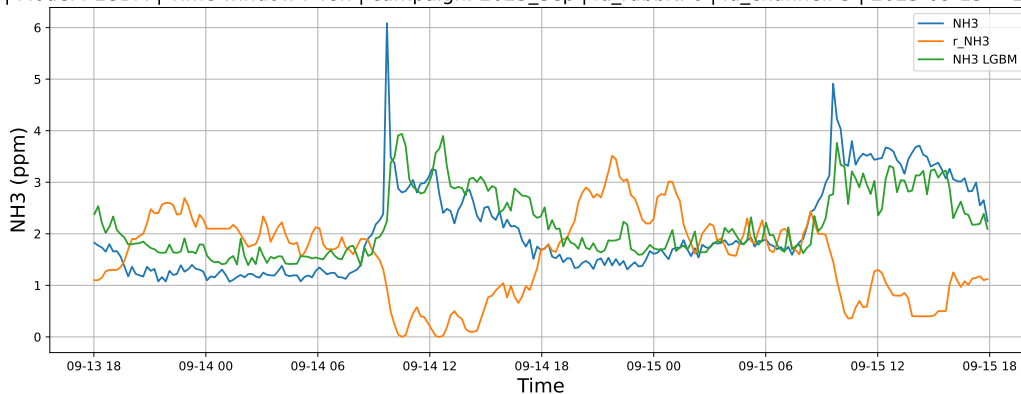


3.1.8.2 Time window : 48

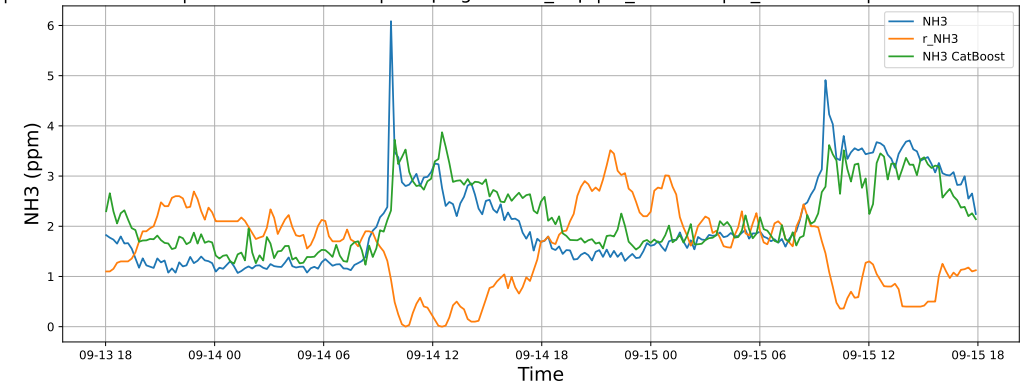
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

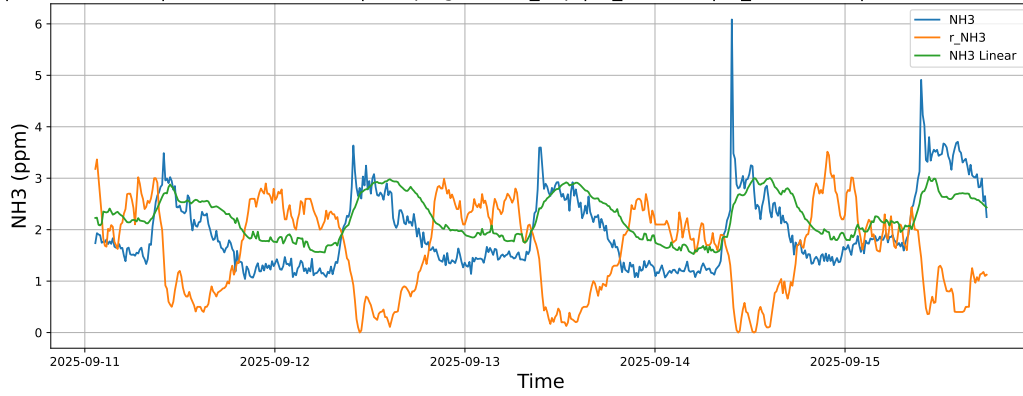


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

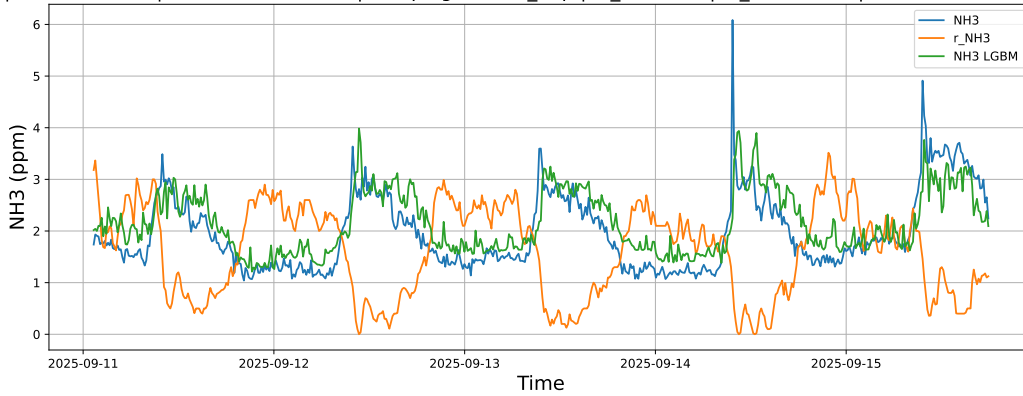


3.1.8.3 Time window : ALL

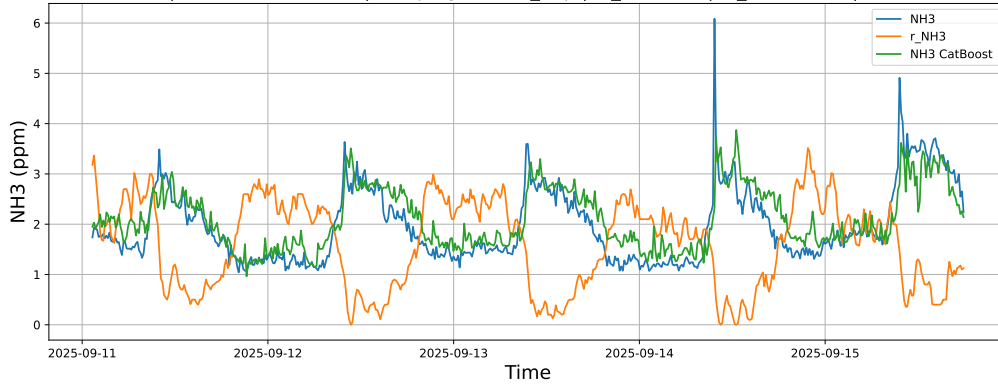
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

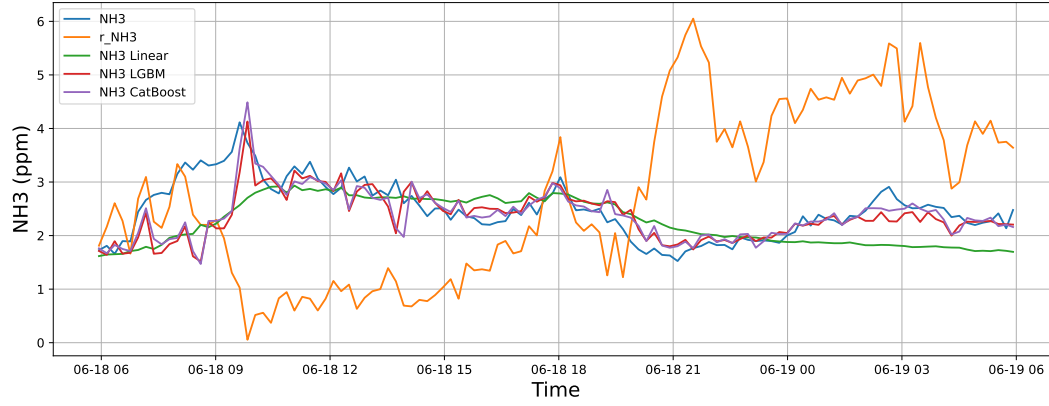


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

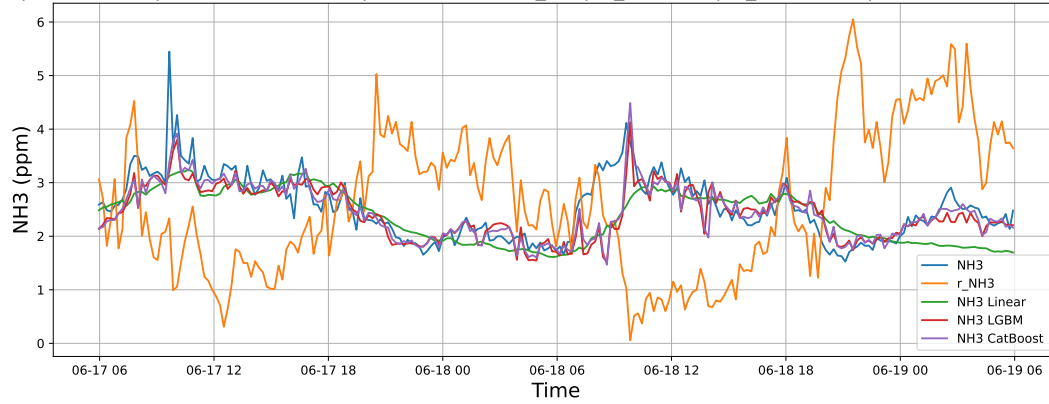
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



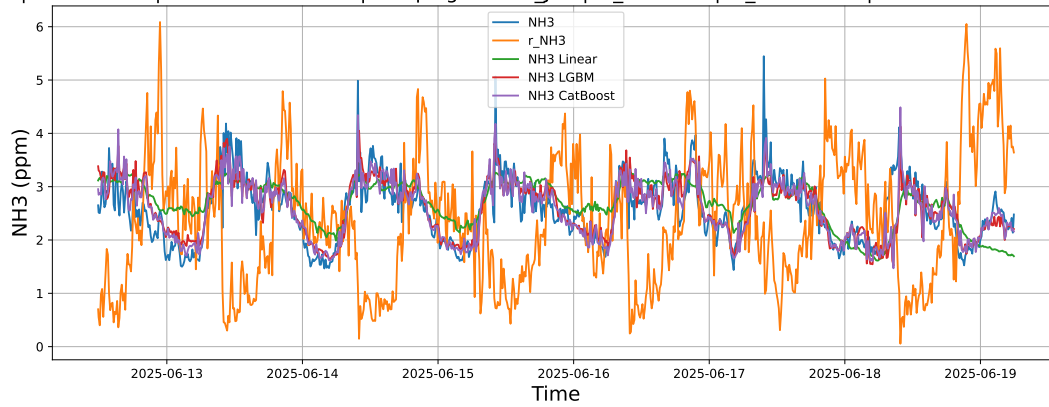
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

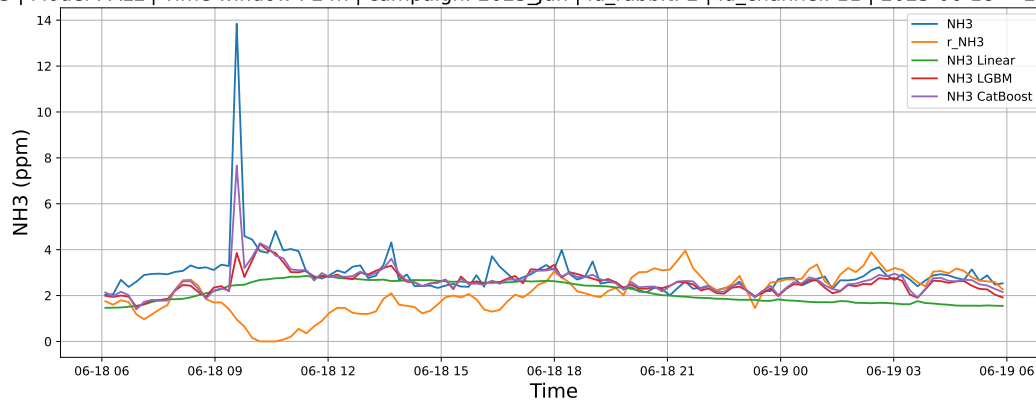
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

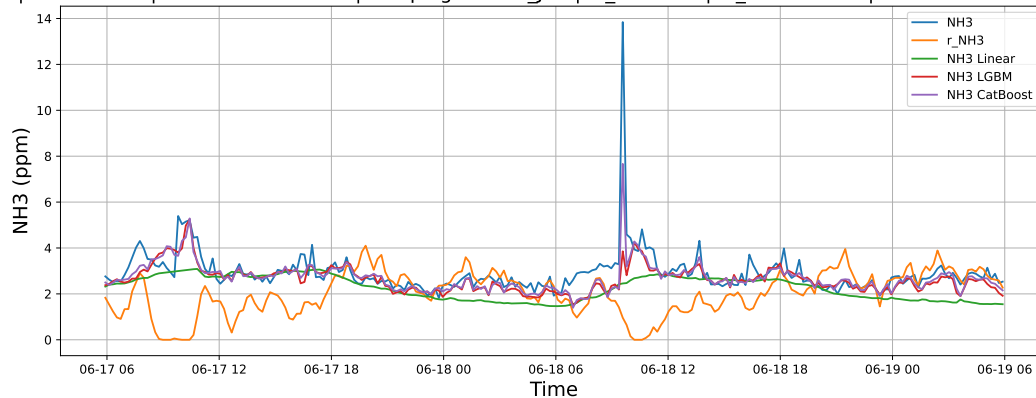
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



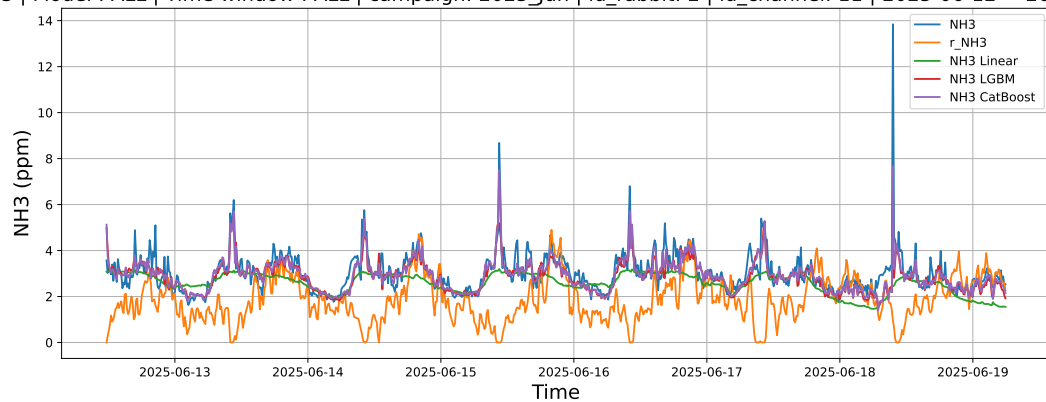
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

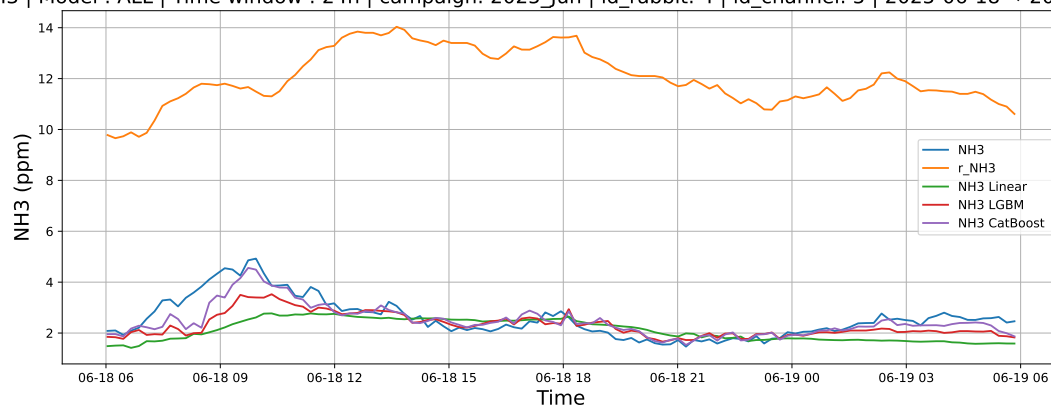
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

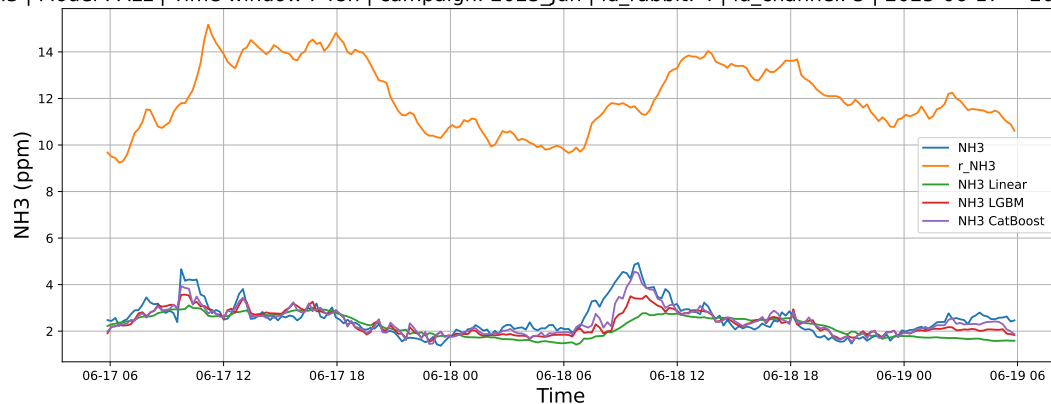
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



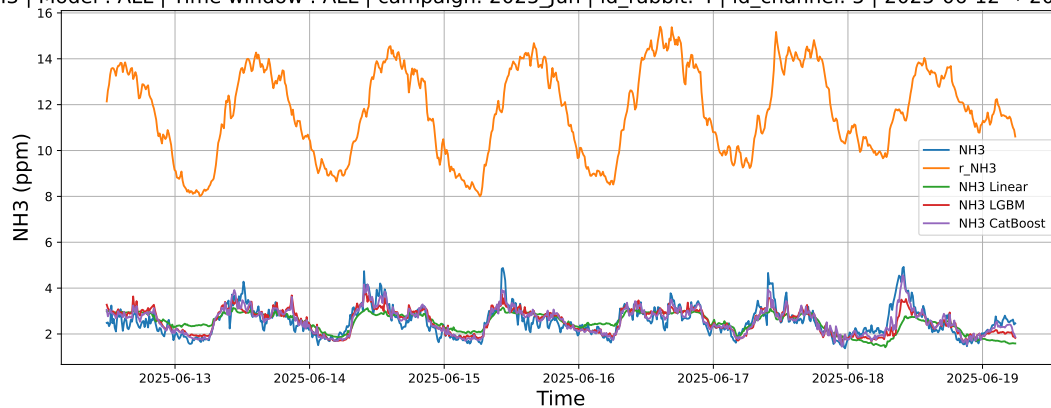
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

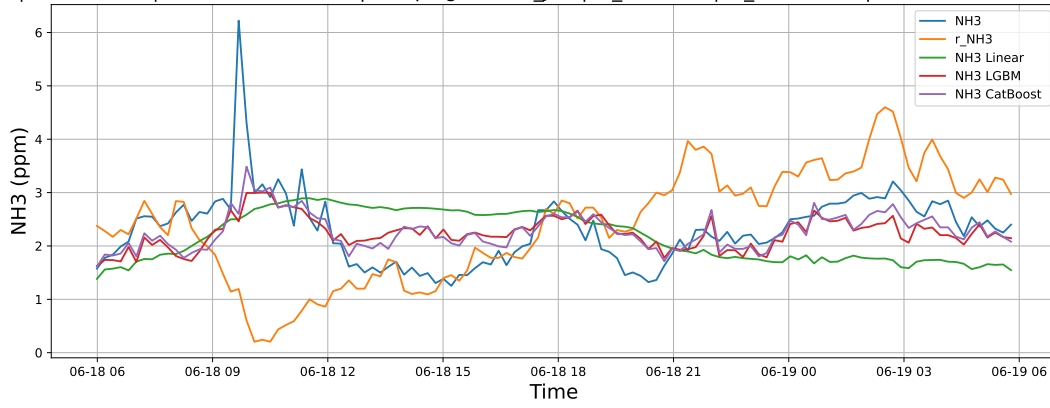
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

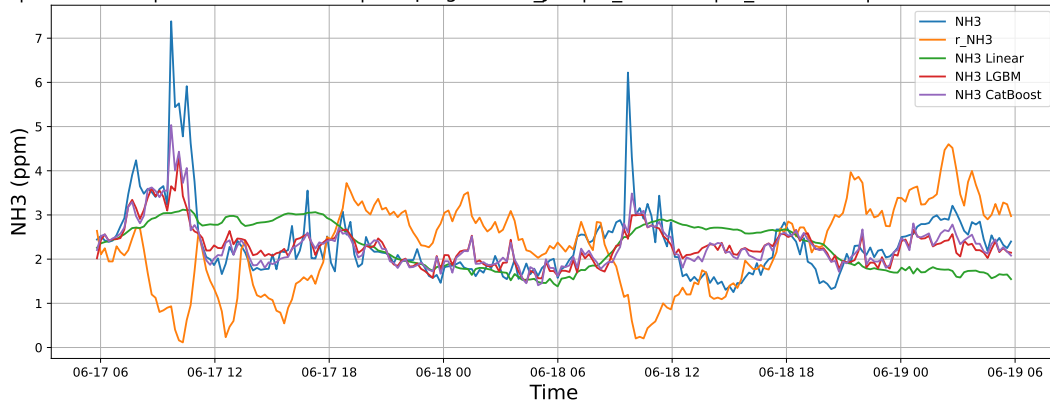
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



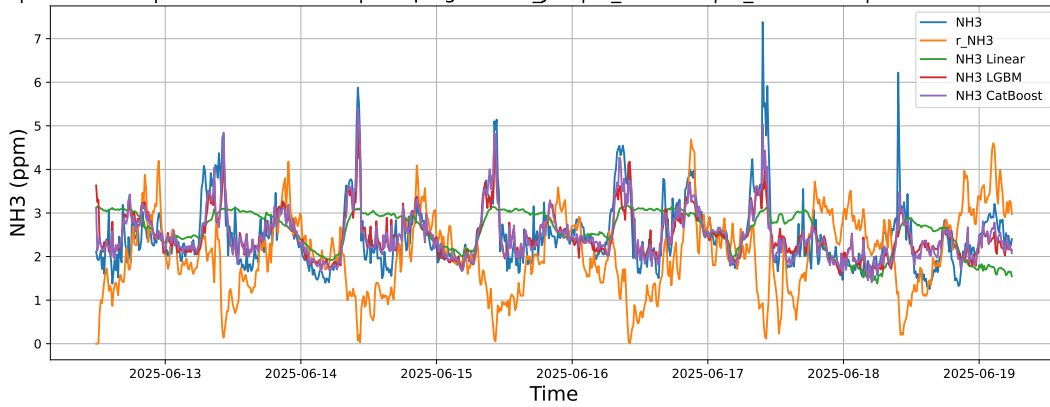
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

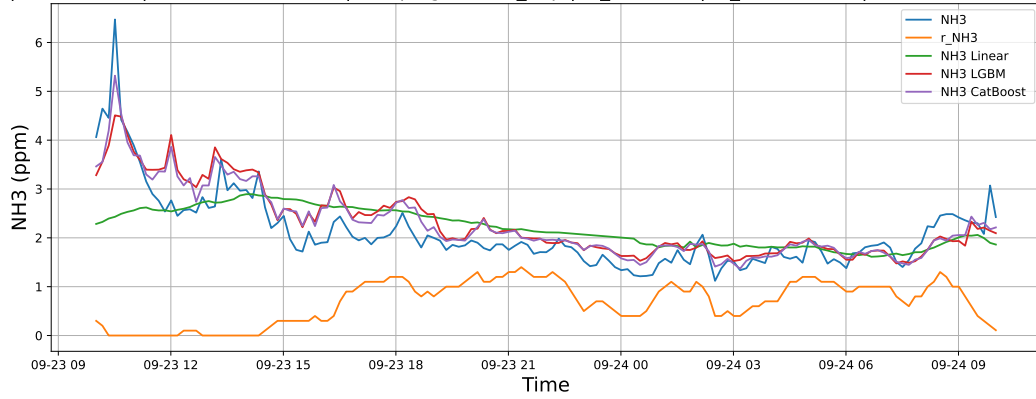
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

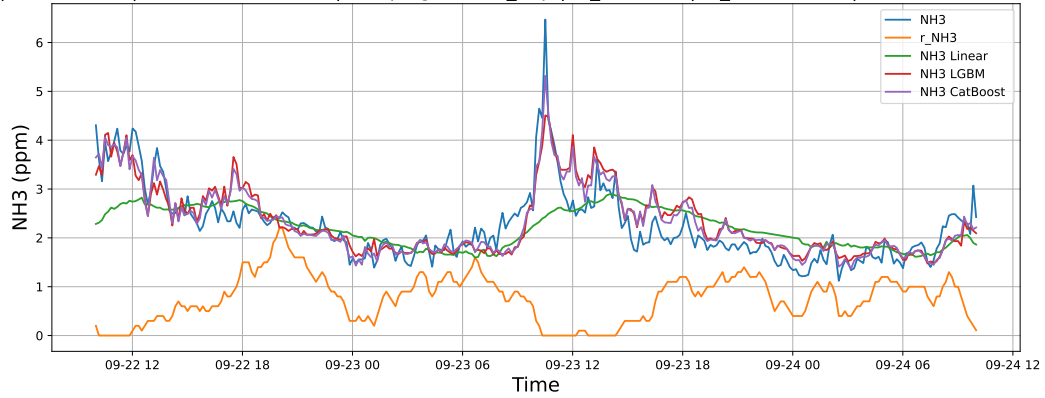
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



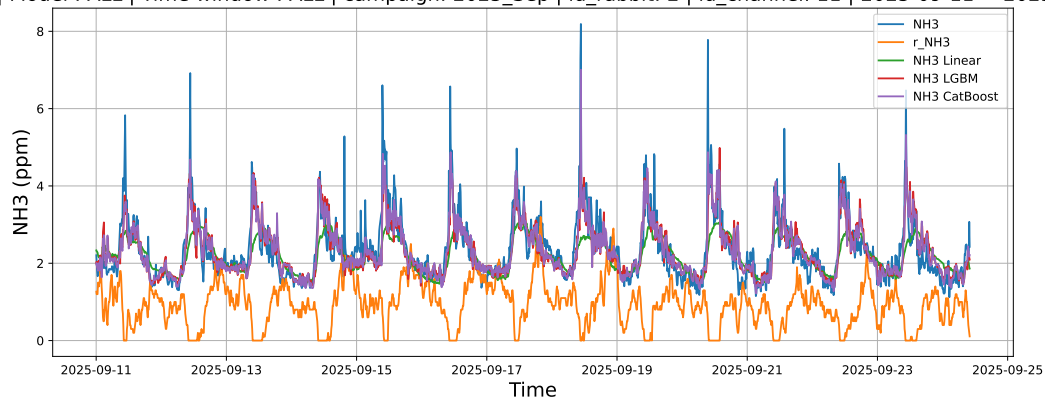
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

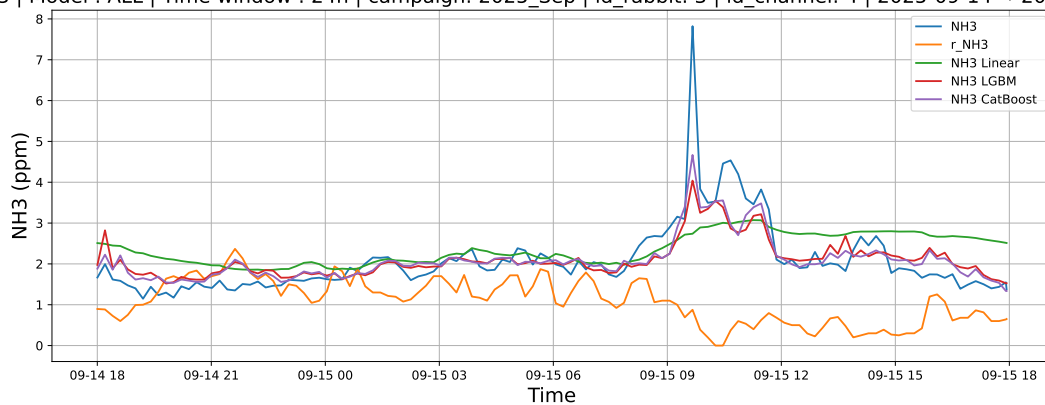
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

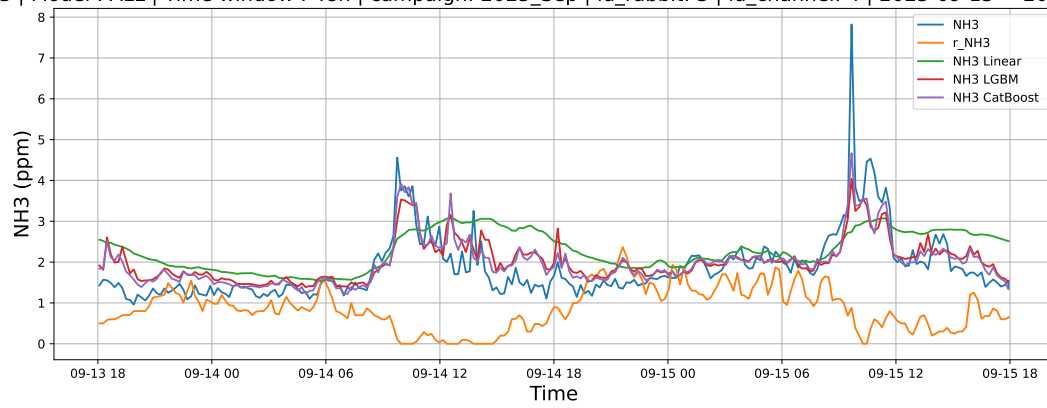
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



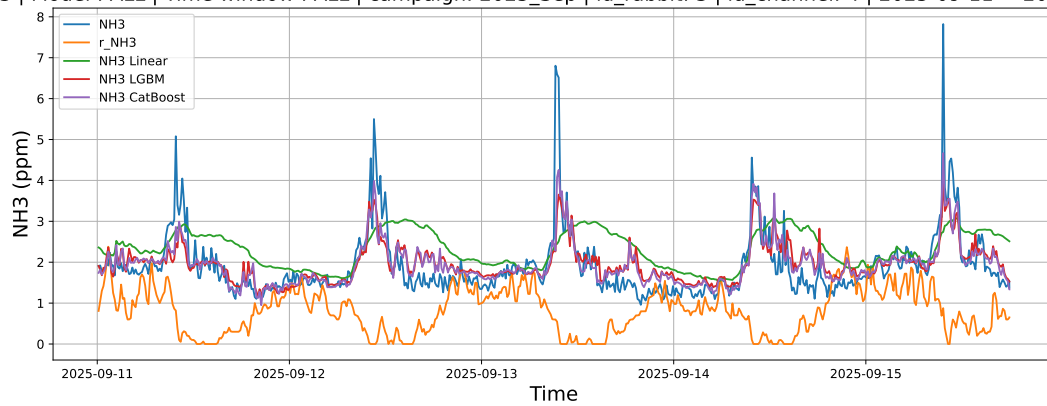
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

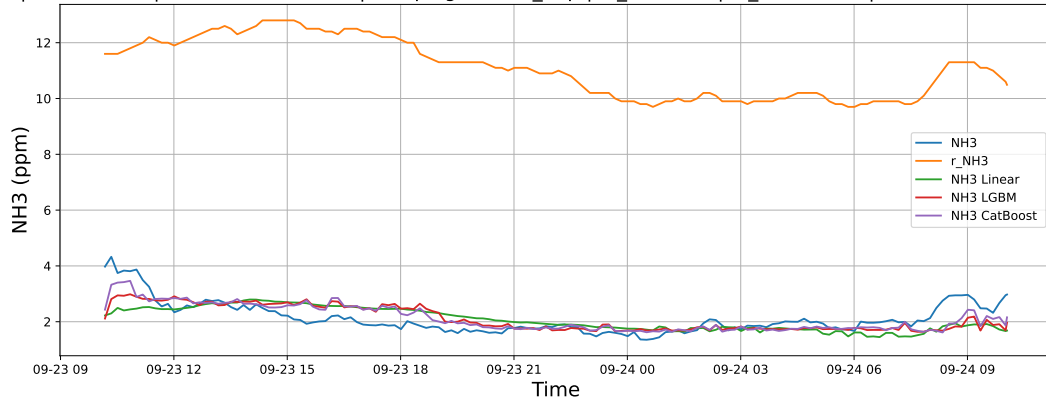
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

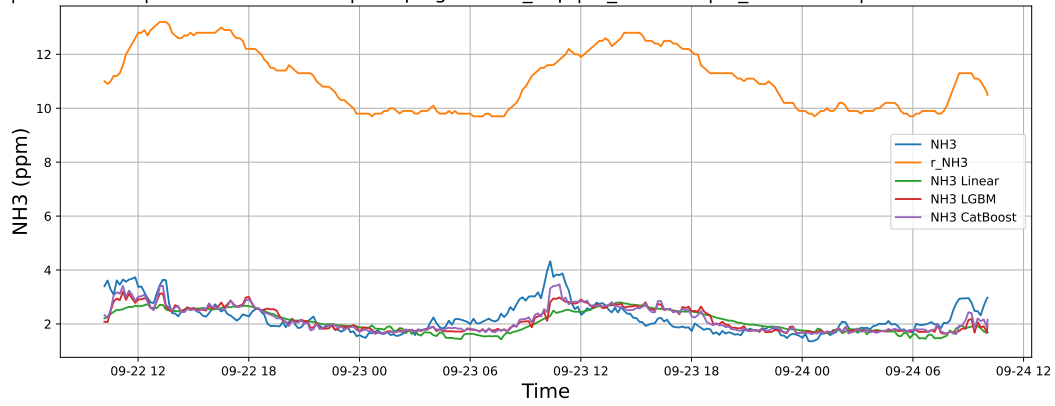
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



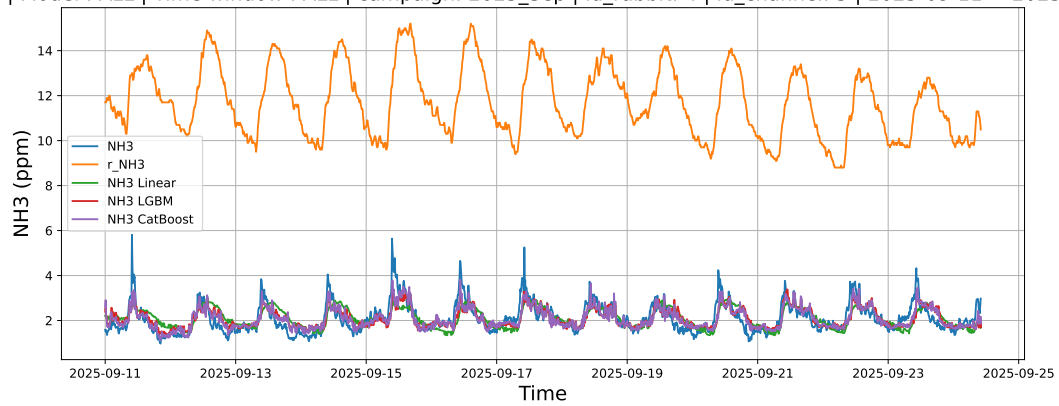
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

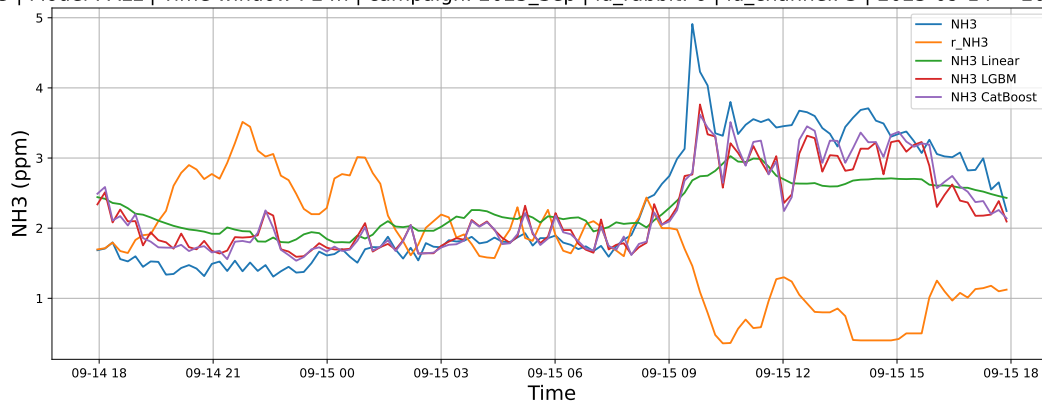
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

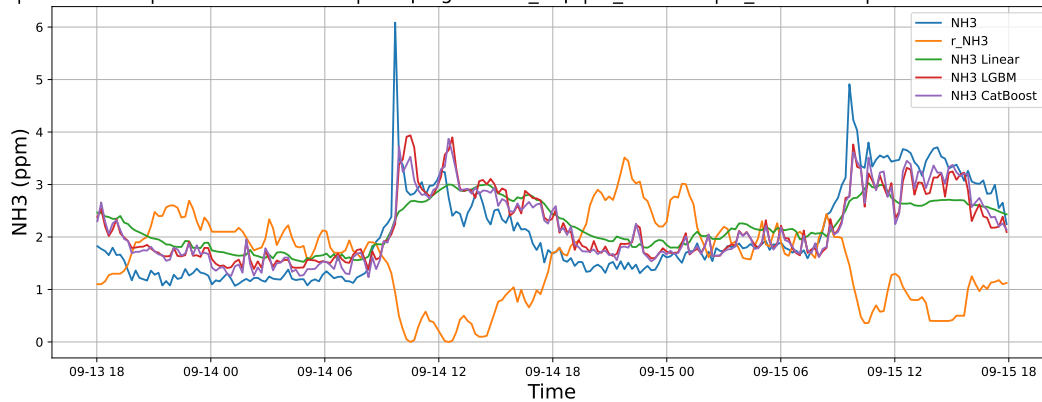
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

